BIF101, Topic 96_105

- **Muhammad Kashif**



Protein Databases

Introduction

Protein databases store

- **Protein sequences**
- **Motif**
- **Structure**
- **Structure alignments**



Protein Databases

Origin

First sequences to be collected were Proteins using **Sanger and Tupy's** methods (1951) where Common protein families like cytochromes were sequenced

Protein Databases

Origin

Atlas of protein sequences (mainly cytochromes) was assembled by Margret Dayhoff and her collaborators at **N**ational **B**iomedical **R**esearch **F**oundation (NBRF) in 1960s

Protein Databases

PIR ((Protein Information Resource)

The collection of Dayhoff and co became **PIR**

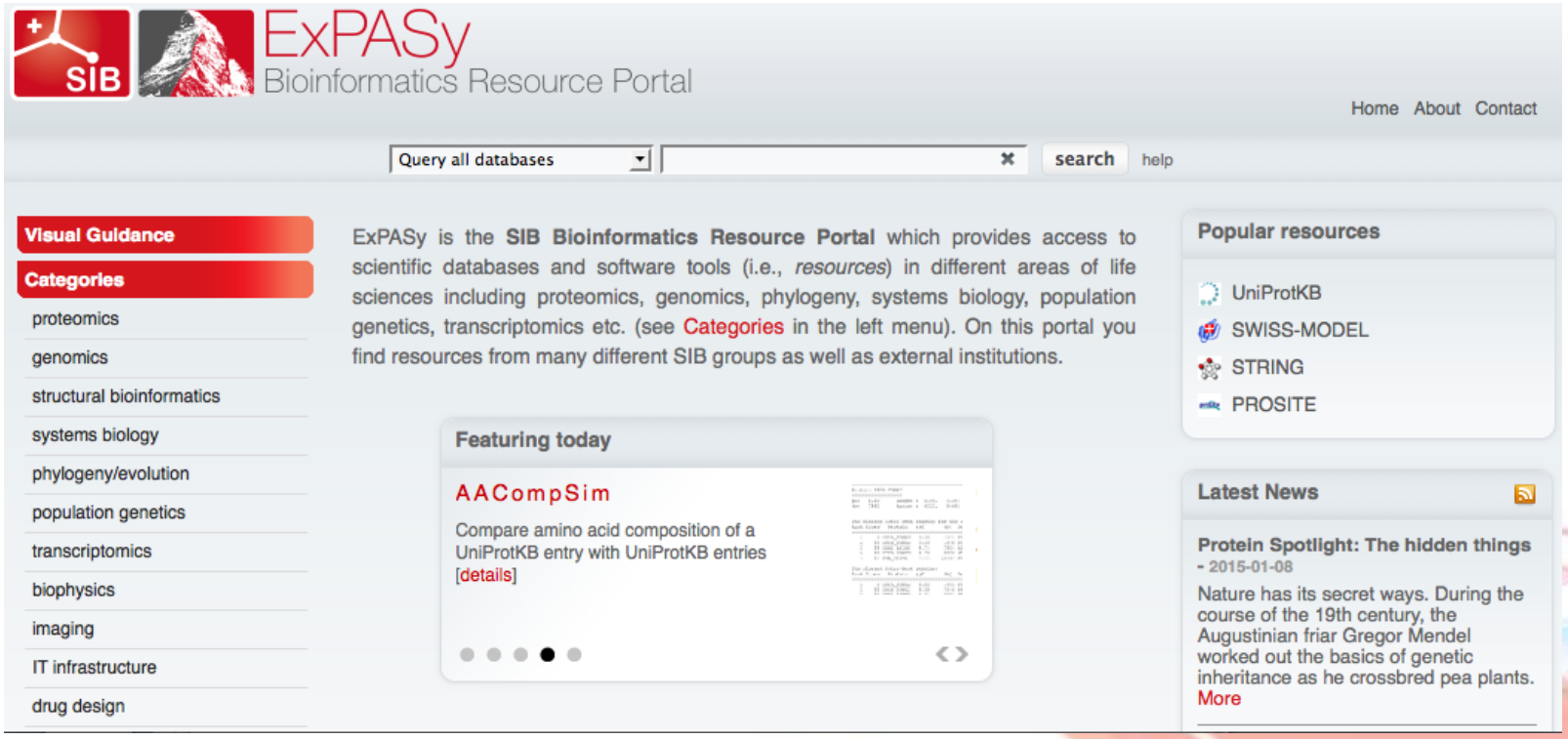
which is now a collaboration of **NBRF**, **M**unich **C**enter for **P**rotein **S**equences (**MIPS**) and **J**apan **I**nternational **P**rotein **I**nformation **D**atabase (**JIPID**)

Protein Databases

Protein Sequences

- **Swiss-Prot** is Collaboration between the SIB and EBI
- Weekly releases from about 50 servers across the world, the main source being ExPASy in Geneva

Protein Databases



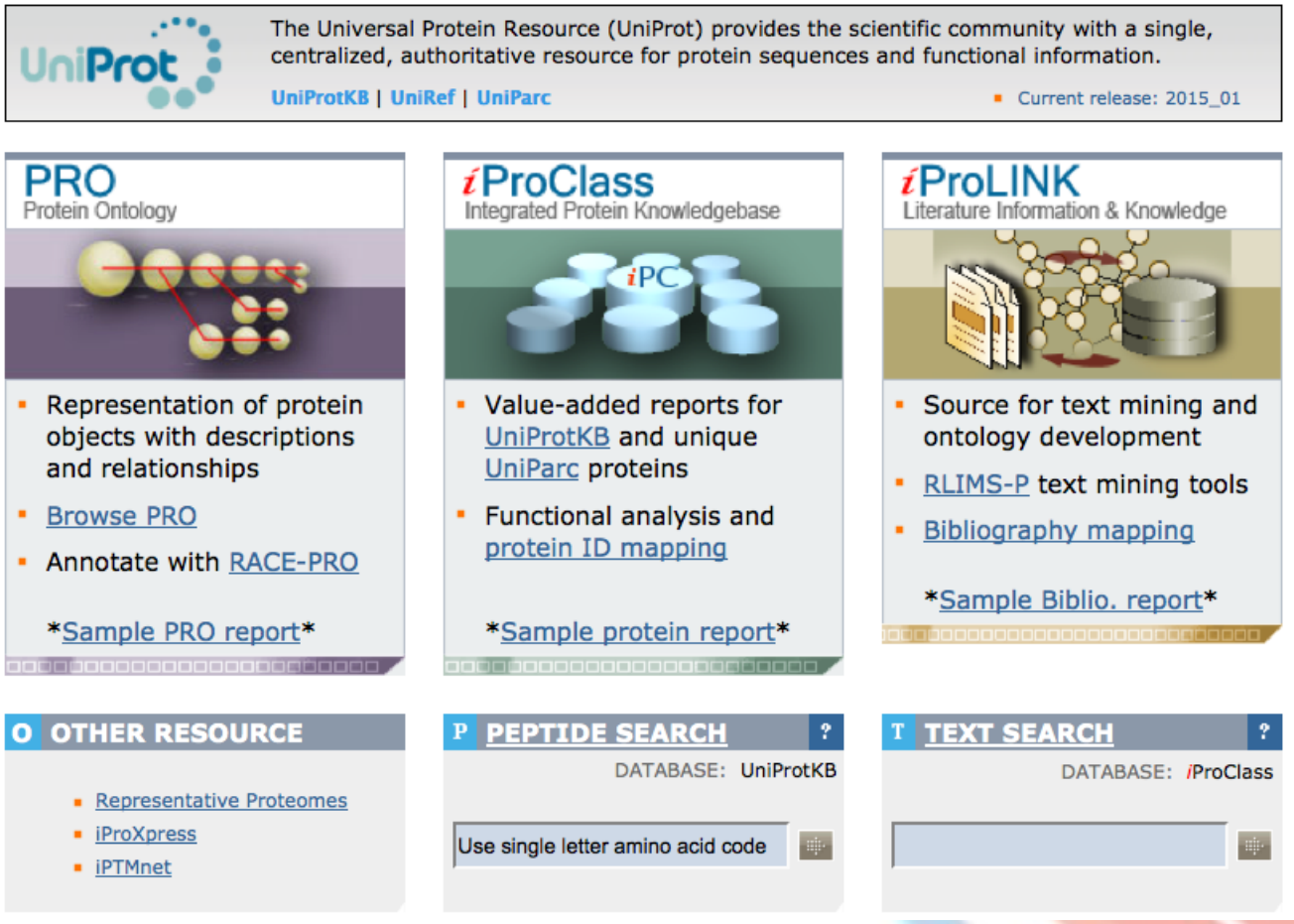
http://www.expasy.org/

Protein Databases

Protein Sequences

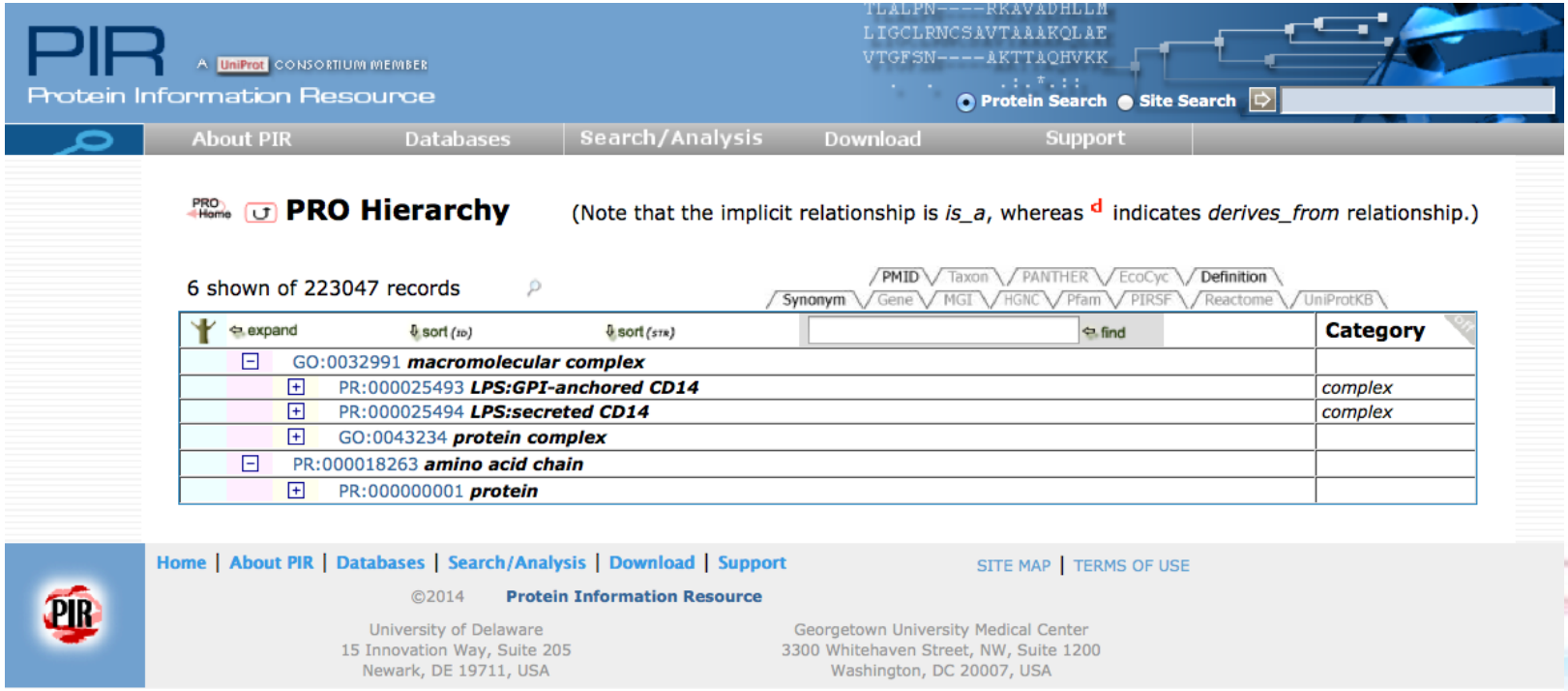
- International partnership between PIR, EBI and SIB
- Created **UniProt**, by unifying the PIR-PSD, Swiss-Prot, and TrEMBL databases

Protein Databases



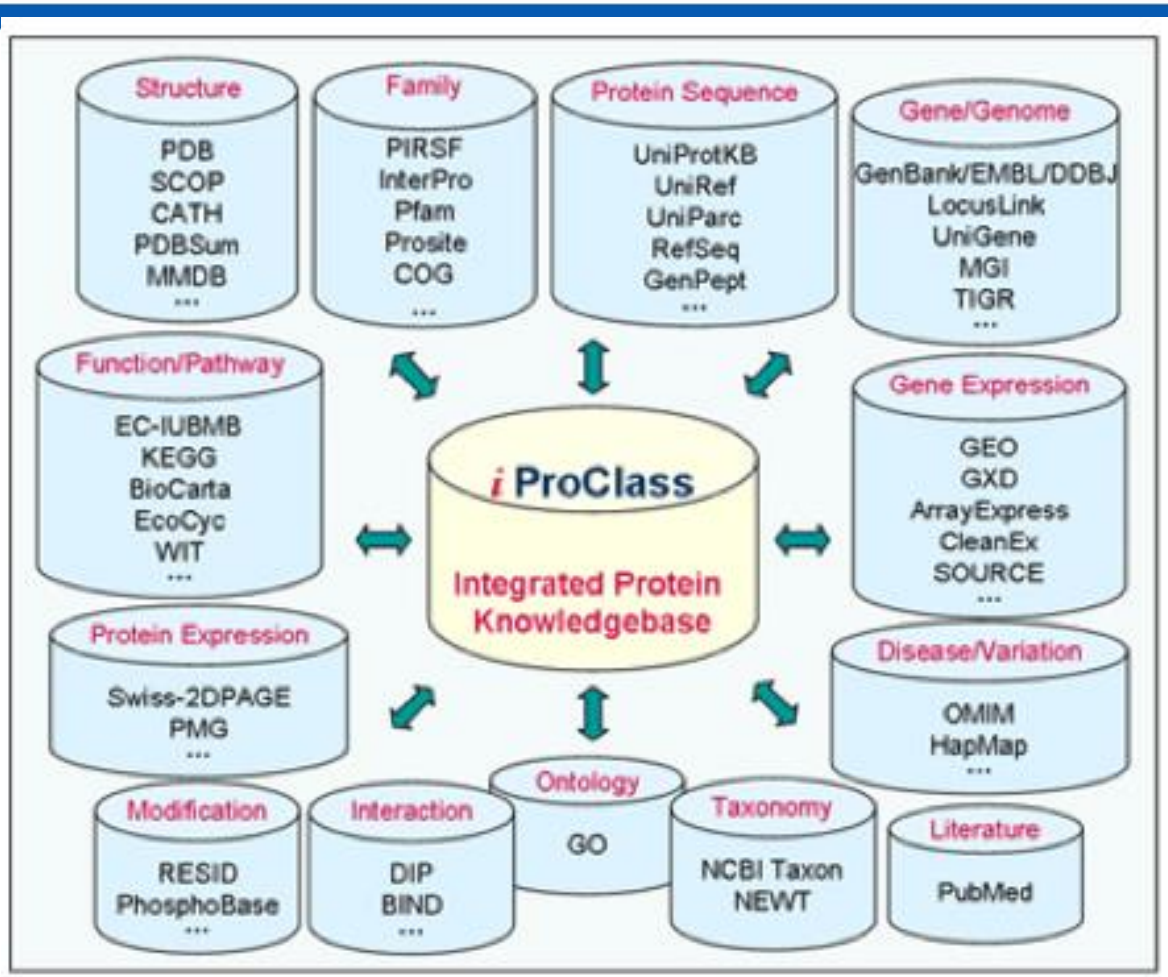
http://www.uniprot.org/

Protein Databases



http://pir.georgetown.edu/

Protein Databases

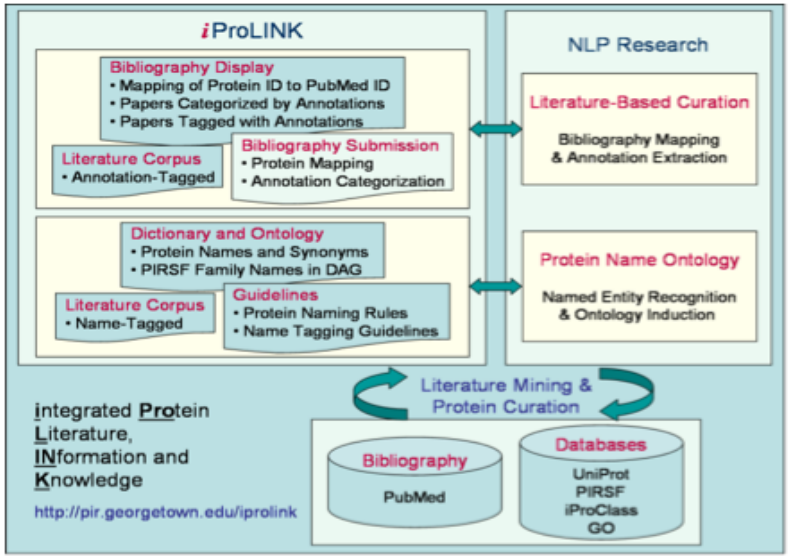


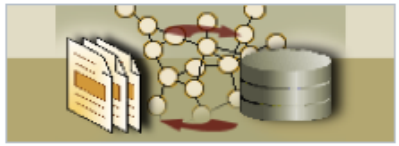
http://www.uniprot.org/

Protein Databases

iProLINK Tutorial

*i*ProLINK (*i*ntegrated **P**rotein **L**iterature, **IN**formation and **K**nowledge) has been developed as a resource to facilitate text mining in the area of literature-based database curation, named entity recognition, and protein ontology development. The collection of data sources can be utilized by computational and biological researchers to explore literature information on proteins and their features or properties (Hu *et al.*, 2004).





Bibliography Mapping/ Annotation Extraction

- Bibliography Display/Submission
- Annotation-Tagged Corpora
- RLIMS-P Text Mining Tool
- eFIP Text Mining Tool
- eGIFT Text Mining Tool
- iSimp Sentence Simplification System

Entity Recognition/ Ontology Development

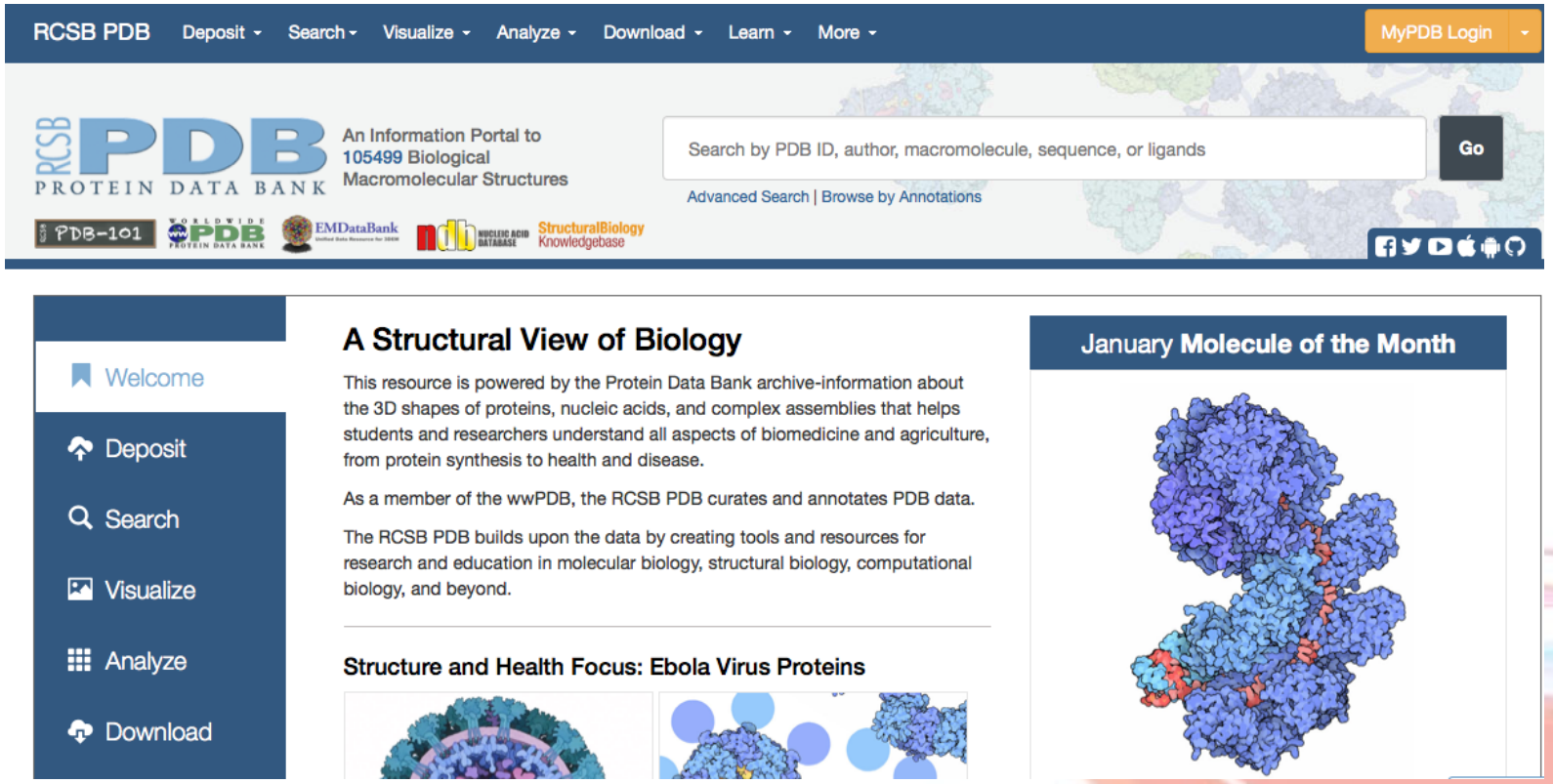
- Name Tagging Guidelines/Corpora
- Protein Ontology Development

Collaborators

***i*ProLINK Paper**

http://www.uniprot.org/

Protein Databases



http://www.rcsb.org/pdb/home/home.do

Protein Databases

SCOPe: Structural Classification of Proteins — extended. Release 2.04 (July 2014, new entries added 2014-12-18)

Browse | Stats & History | ASTRAL Subsets | Downloads | Related Resources | References | Help | About

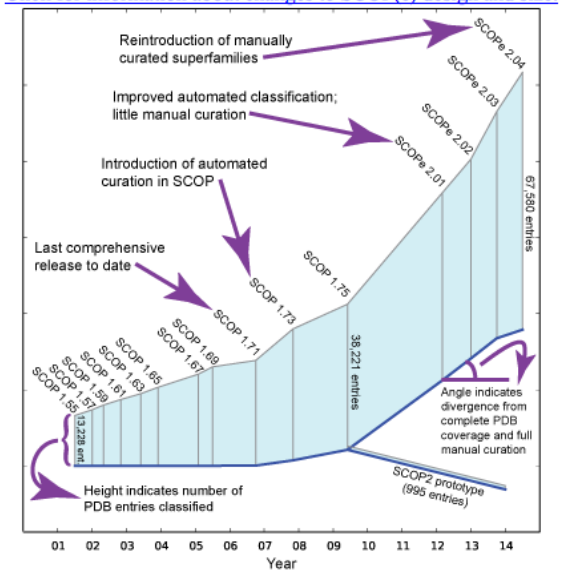
Welcome to SCOPe!
SCOPe is a database developed at the Berkeley Lab and UC Berkeley to extend the development and maintenance of SCOP.
SCOP was conceived at the MRC Laboratory of Molecular Biology, and developed in collaboration with researchers in Berkeley.
Work on SCOP (version 1) concluded in June 2009 with the release of SCOP 1.75.
SCOPe classifies many newer structures through a combination of automation and manual curation, and corrects some errors in SCOP,
aiming to have the same accuracy as the hand-curated SCOP releases. **SCOPe** also incorporates and updates the ASTRAL database.
For prior releases, click on the Stats & History tab above. For more info, click on the About tab above.
New PDB entries were last added on **2014-12-18**; for more info on periodic updates click on the Help tab above.

Search **SCOPe** (example): Search

Classes in SCOPe 2.04:

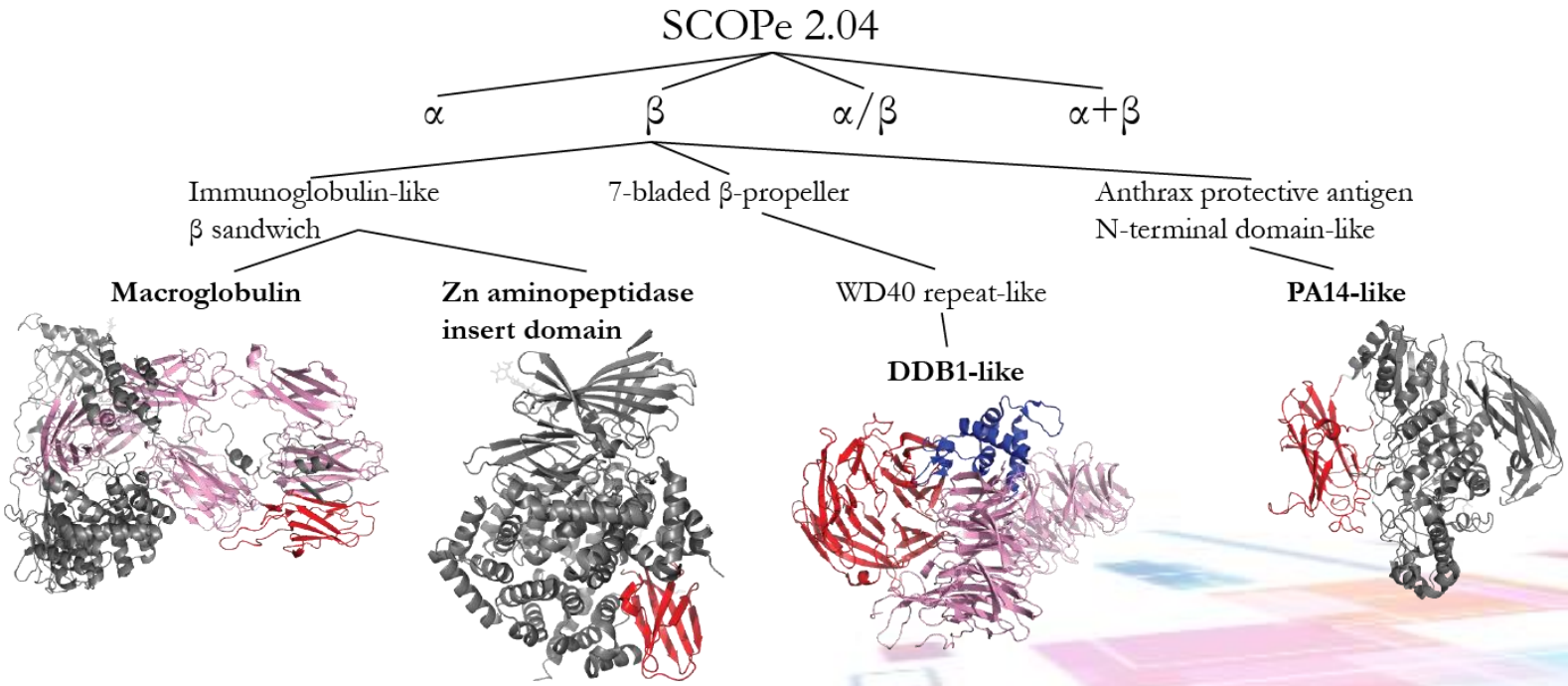
1. a: All alpha proteins [46456] (285 folds)
2. b: All beta proteins [48724] (176 folds)
3. c: Alpha and beta proteins (a/b) [51349] (148 folds)
4. d: Alpha and beta proteins (a+b) [53931] (380 folds)
5. e: Multi-domain proteins (alpha and beta) [56572] (68 folds)
6. f: Membrane and cell surface proteins and peptides [56835] (57 folds)
7. g: Small proteins [56992] (91 folds)
8. h: Coiled coil proteins [57942] (7 folds)
9. i: Low resolution protein structures [58117] (25 folds)

Click for information about changes to SCOP(e) design and size.



http://scop.mrc-lmb.cam.ac.uk/scop/

Protein Databases



http://scop.mrc-lmb.cam.ac.uk/scop/

Protein Databases

Conclusions

- First sequences to be collected were Protein sequences
- Protein databases are classified on the basis of sequences, motifs and structures
- Growth of Sequence in Databases is exponential

Genome & Organism Specific Databases

Origin

- First attempt to sequence free living organism was launched in late 1990's
  - (Blattner et al. 1997)
- Viruses had already been sequenced

Genome & Organism Specific Databases

Origin

- ***Haemophilus influenzae*** was the first published genome
  - (Fleischmann et al. 1995)
- The project initiated at **T**he **I**nstitute of **G**enome **R**esearch **(TIGR)** under leadership of **Craig Ventor**

Genome & Organism Specific Databases

Origin

- Use of **shotgun sequencing method** speeded up the process
- 1.8 million bp
- Took 9-months and cost was 1 million
- Paved the way for sequencing of many other organisms

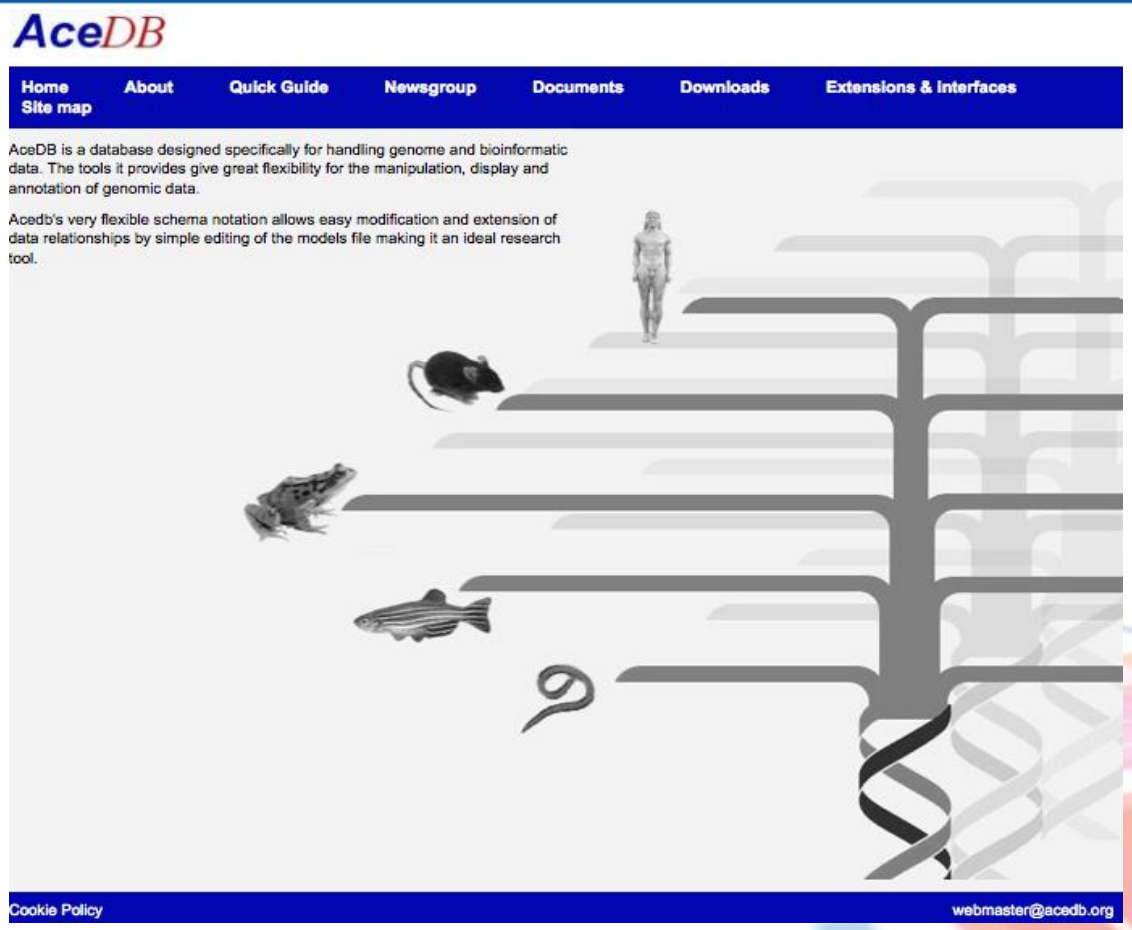
Genome & Organism Specific Databases

Examples

- **AceDB** (**A *C. e****legans*** **D**ata**B**ase) was the first genome database developed in 1989
  - by **Richard durbin** and **Thierry-Miegi**



Genome & Organism Specific Databases



http://www.acedb.org/

Genome & Organism Specific Databases

Examples

- **TAIR** (**T**he **A**rabidopsis **I**nformation **R**esource) http://www.arabidopsis.org/ and **SGB** (**S**accharomyces **G**enome **D**atabase) http://www.yeastgenome.org/ utilized **AceDB** system

Genome & Organism Specific Databases

Human Genome Project

- Pilot project, the **Human Genome Initiative** begun by **D**epartment **O**f **E**nergy (**DOE**) in 1986

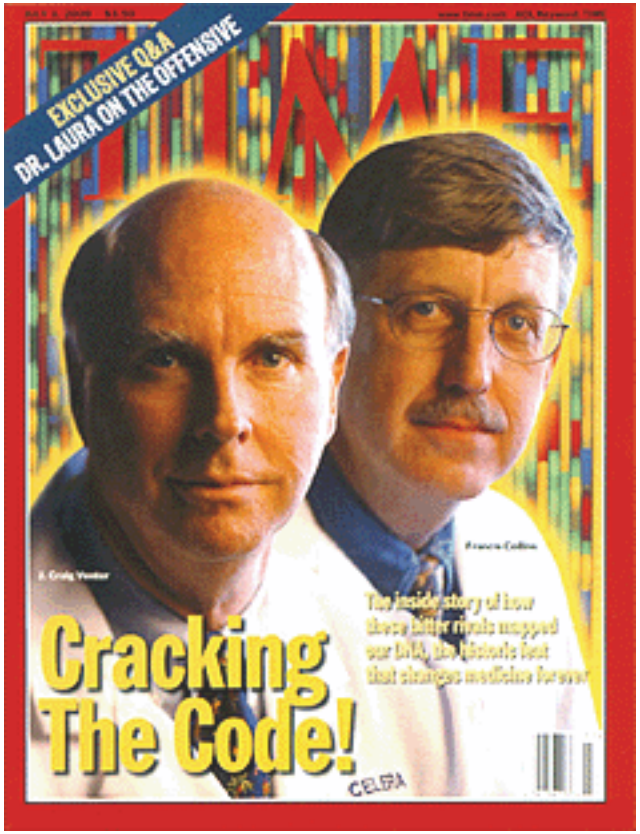


Genome & Organism Specific Databases

Human Genome Project

- **N**ational **H**uman **G**enome **R**esearch **I**nitiative (**NHGRI**), a federally funded organization (NIH) started in 1988 **(Francis Collin)**
- Celera, a commercial vendor **(Craig Venter)** joined the venture in 1998

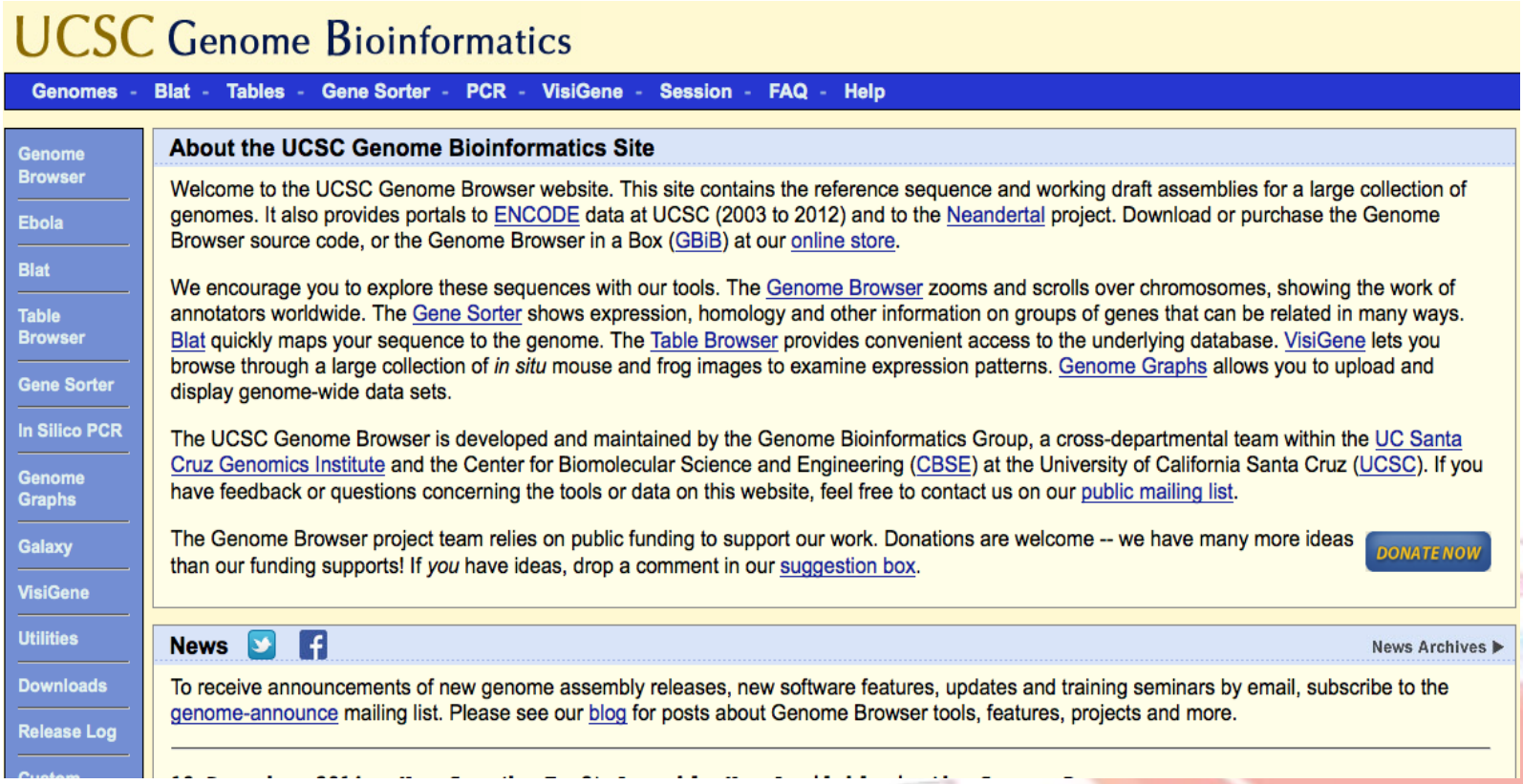
Genome & Organism Specific Databases



Human Genome Project

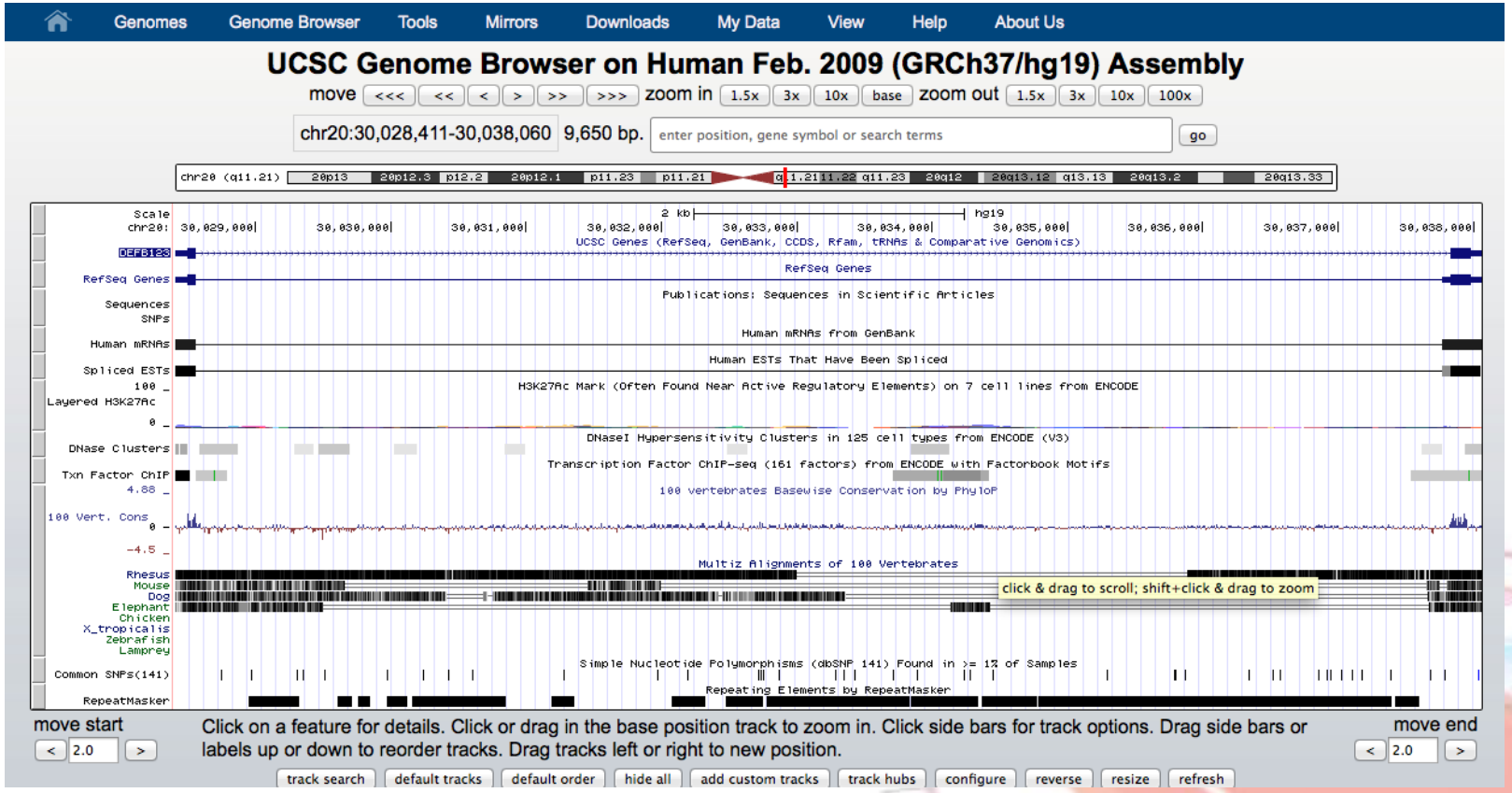
- Both simultaneously announced the completion of the project in 2000
- Total 3.4 billion bases sequenced at a cost of $1/base

Genome & Organism Specific Databases



http://genome.ucsc.edu/

Genome & Organism Specific Databases



http://genome.ucsc.edu/

Genome & Organism Specific Databases

Conclusion

- Success of ***Haemophilus influenzae*** paved the way for other genome sequencing projects
- Human Genome Project was accomplished by NHGRI and Celera
- Genome browsers help viewing Genome features

Gene Expression Databases

Gene Expression Omnibus (GEO)

- A public repository for the archiving and distribution of gene expression data submitted by the scientific community

Gene Expression Databases

Gene Expression Omnibus (GEO)

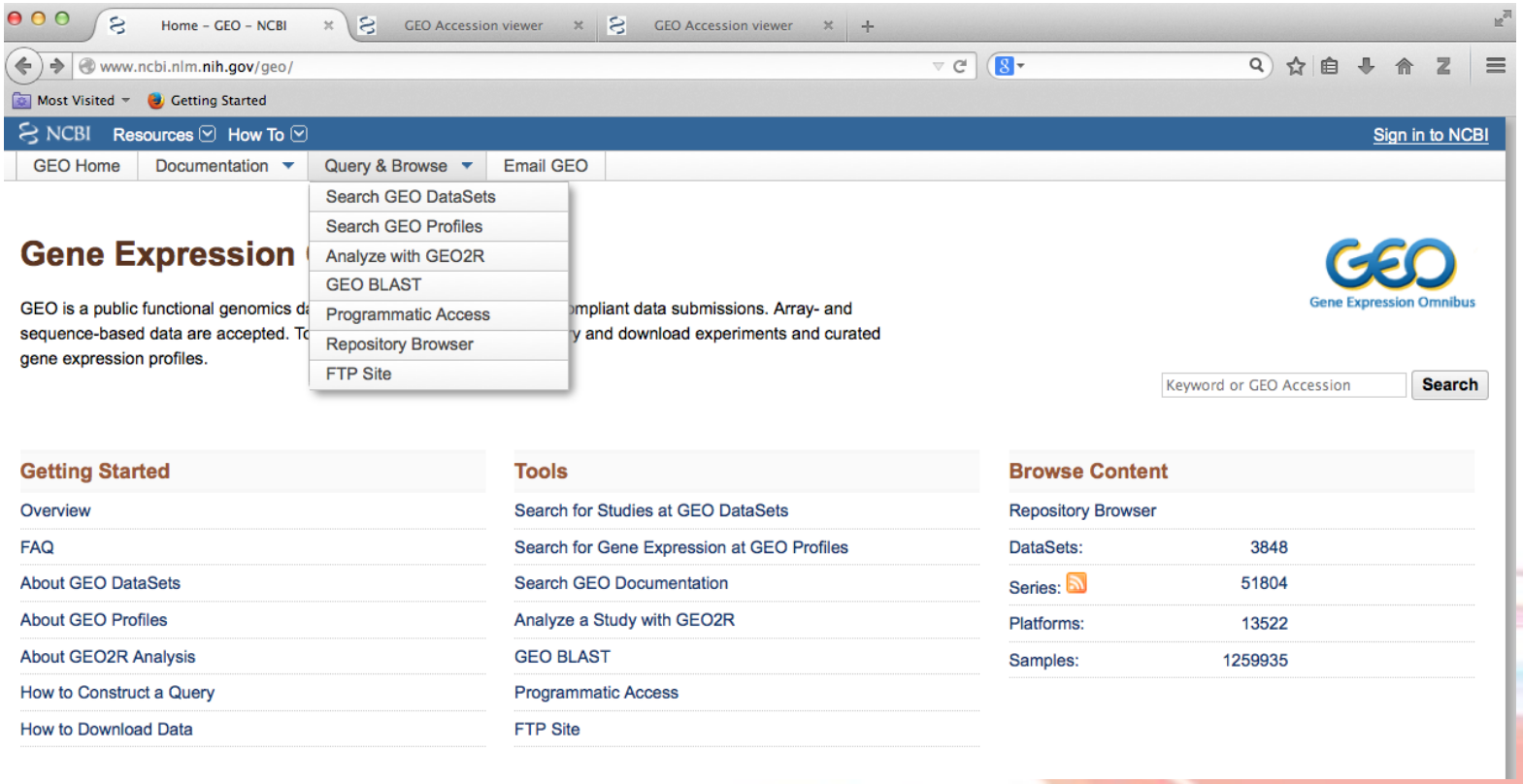
- MIAME compliant data
  - Minimum Information About a Microarray Experiment http://www.mged.org/Workgroups/MIAME/miame.html

Gene Expression Databases

Gene Expression Omnibus (GEO)

- Convenient for deposition of gene expression data, as required by funding agencies and journals
- Curated resource for gene expression data
- Browsing, querying, analysis and retrieval

Gene Expression Databases



http://www.ncbi.nlm.nih.gov/geo/

Gene Expression Databases

Gene Architecture

GEO has four kinds of records

- **Sample (GSM)**
  preparation and description of the samples
- **Platform (GPL)**
  technology used and the features detected
  - microarray or RNASeq

Gene Expression Databases

Gene Architecture

GEO has four kinds of records

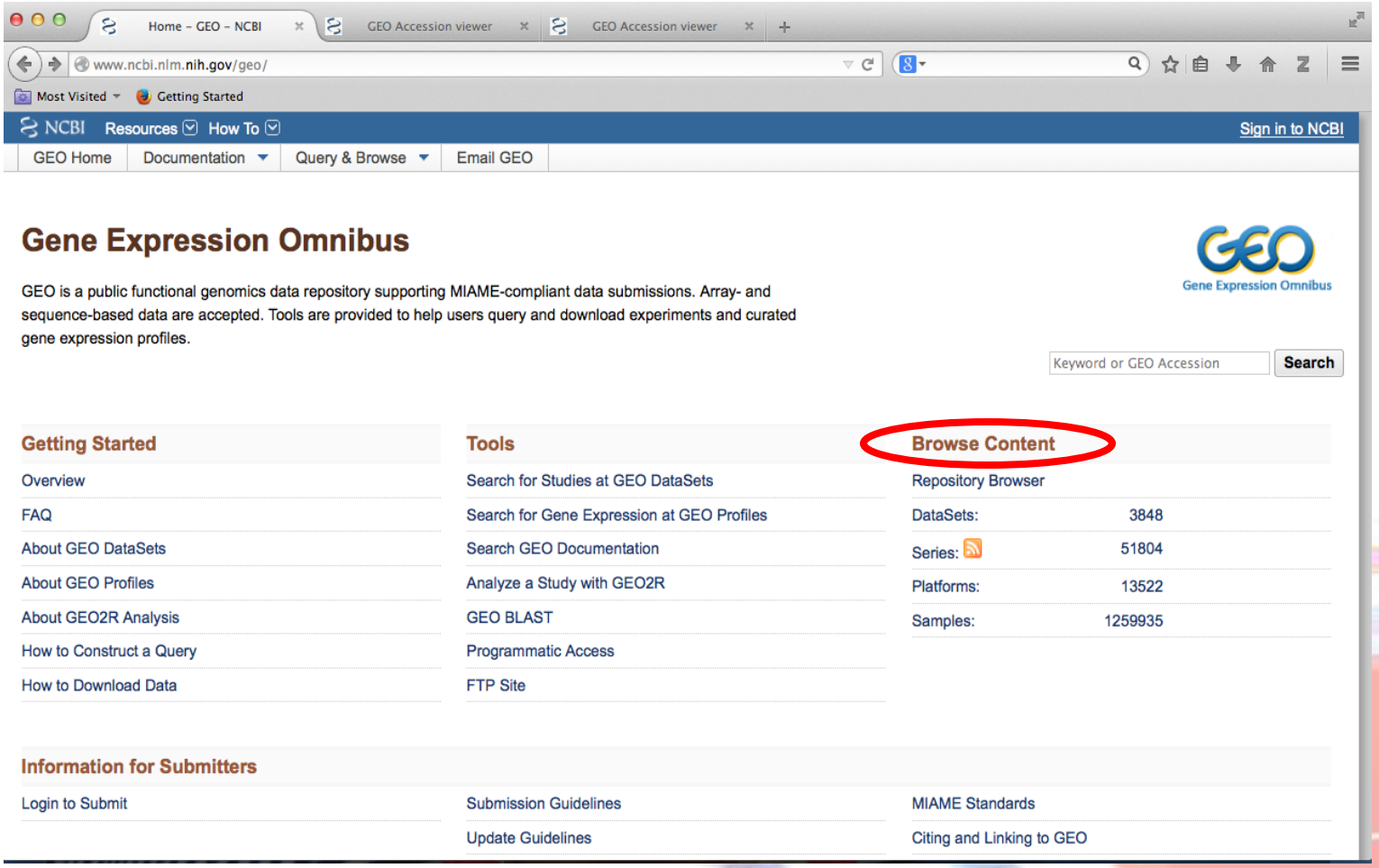
- **Series (GSE)**

  defines a set of samples and how they are related

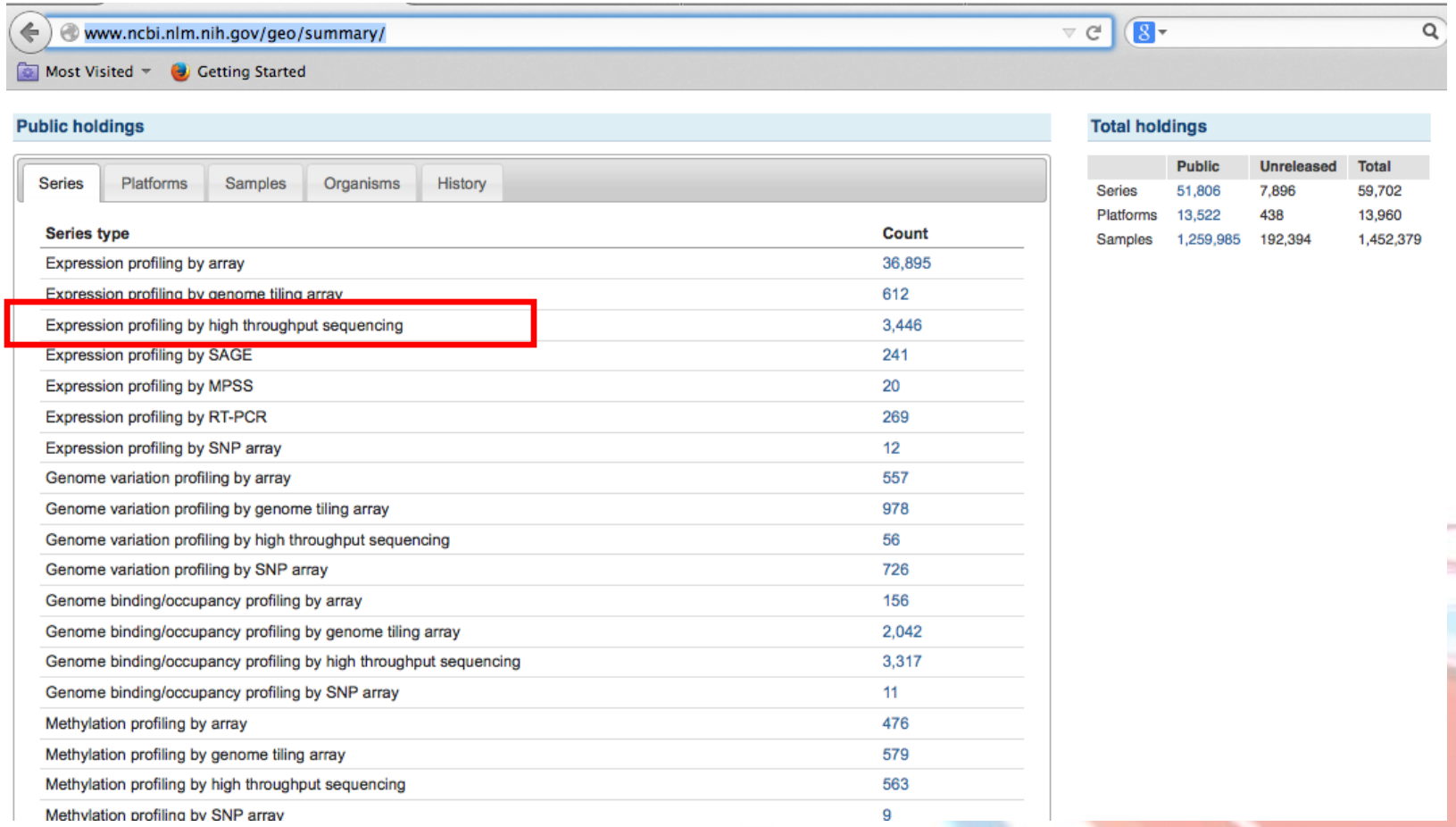
- **Datasets (GDS)**

  sample data collections assembled by GEO

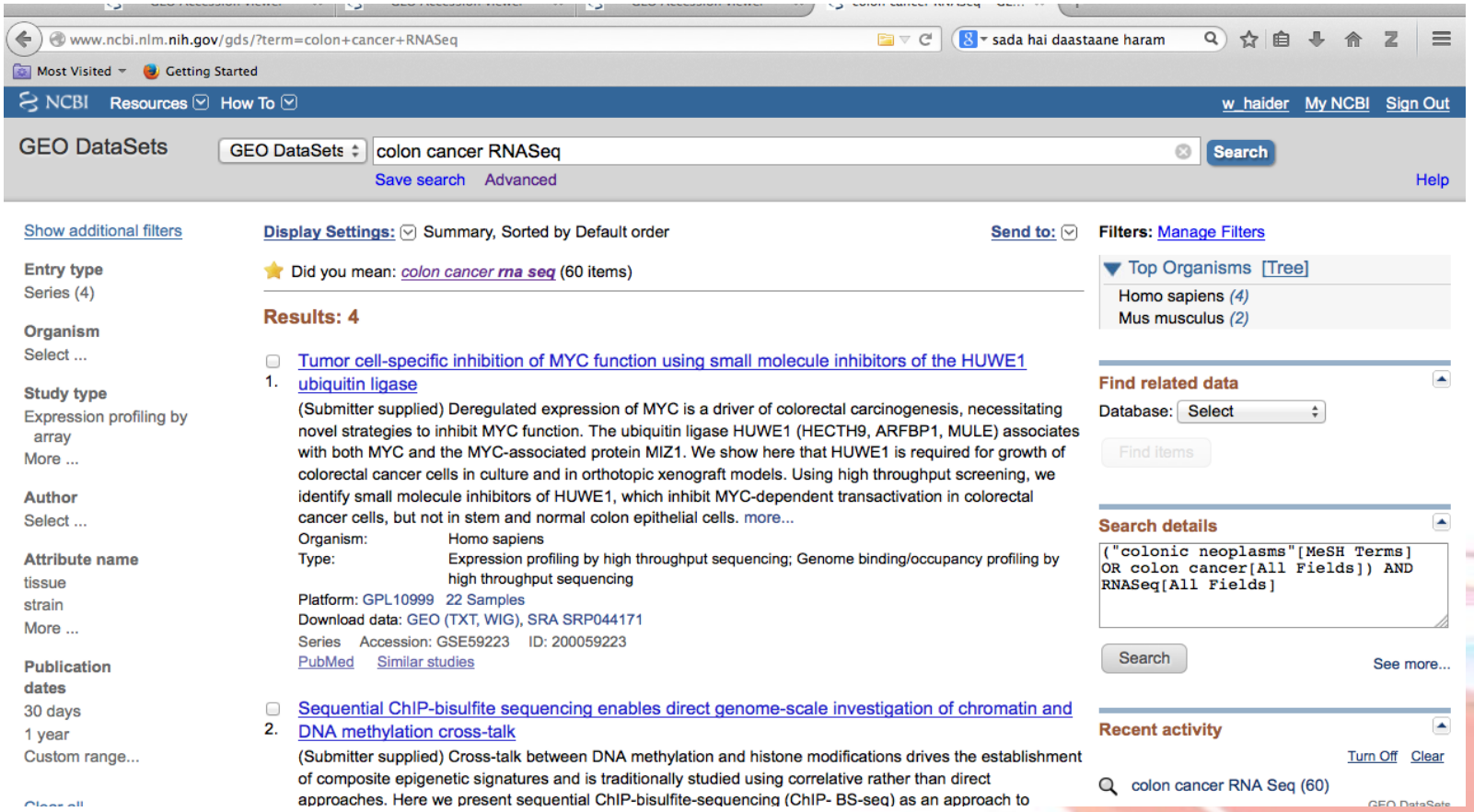
Gene Expression Databases



Gene Expression Databases



Gene Expression Databases

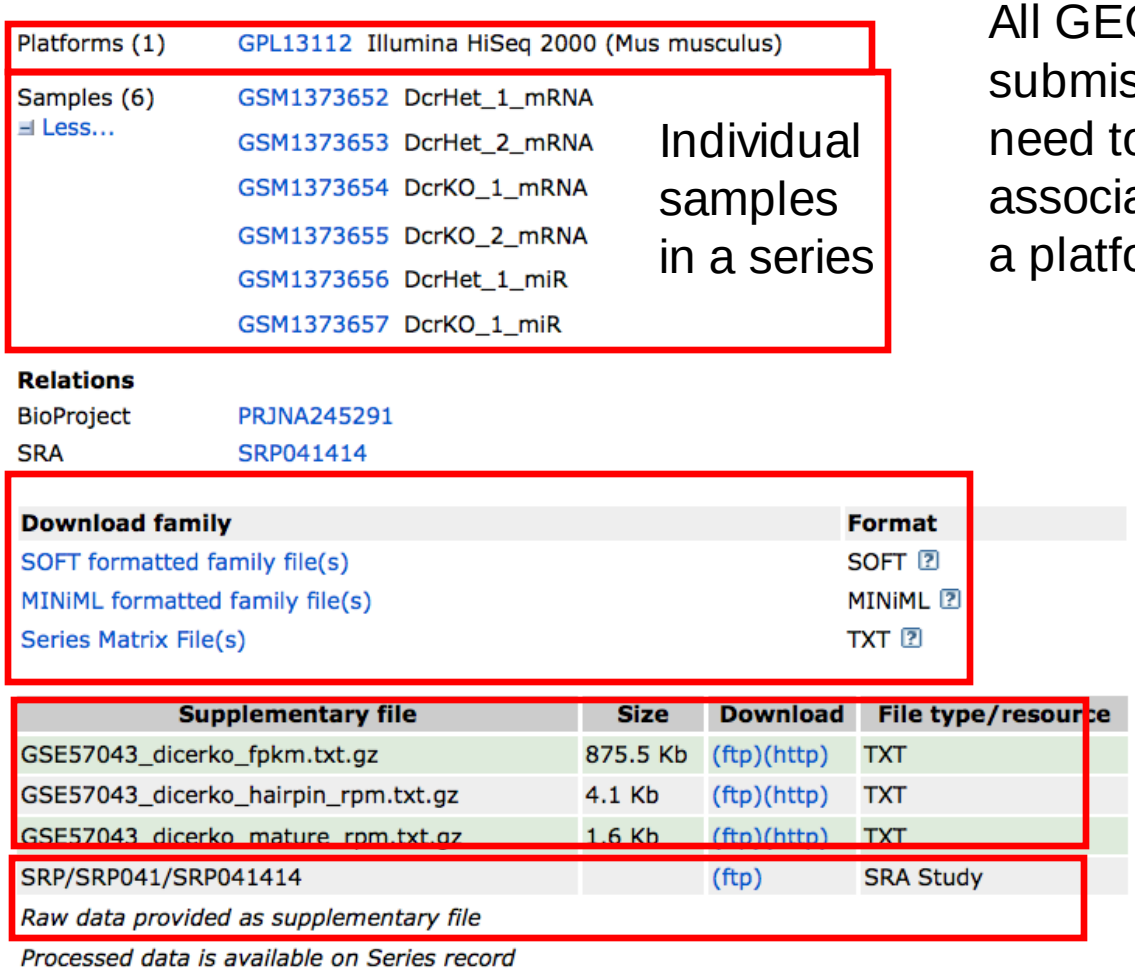


Gene Expression Databases



Data is submitted to GEO as a Series, which represents the experiment design

Gene Expression Databases



Platforms (1)	GPL13112 Illumina HiSeq 2000 (Mus musculus)	
Samples (6) Less...	GSM1373652 DcrHet_1_mRNA	Individual samples in a series
	GSM1373653 DcrHet_2_mRNA	
	GSM1373654 DcrKO_1_mRNA	
	GSM1373655 DcrKO_2_mRNA	
	GSM1373656 DcrHet_1_miR	
	GSM1373657 DcrKO_1_miR	

Relations

BioProject	PRJNA245291
SRA	SRP041414

Download family	Format
SOFT formatted family file(s)	SOFT
MINiML formatted family file(s)	MINiML
Series Matrix File(s)	TXT

Supplementary file	Size	Download	File type/resource
GSE57043_dicerko_fpkm.txt.gz	875.5 Kb	(ftp)(http)	TXT
GSE57043_dicerko_hairpin_rpm.txt.gz	4.1 Kb	(ftp)(http)	TXT
GSE57043_dicerko_mature_rpm.txt.gz	1.6 Kb	(ftp)(http)	TXT
SRP/SRP041/SRP041414		(ftp)	SRA Study

Raw data provided as supplementary file

Processed data is available on Series record

All GEO submissions need to be associated with a platform file.

Click ? One by one

Normalized counts

Raw Reads

Gene Expression Databases

PNAS

An expression signature for p53 status in human breast cancer predicts mutation status, transcriptional effects, and patient survival

Lance D. Miller*†, Johanna Smeds‡, Joshy George*, Vinsensius B. Vega*, Liza Vergara*, Alexander Ploner§, Yudi Pawitan§, Per Hall§, Sigrid Klaar‡, Edison T. Liu*†, and Jonas Bergh‡

*Genome Institute of Singapore, 60 Biopolis Street, #02-01, Singapore 138672; ‡Department of Oncology and Pathology, Radiumhemmet, Karolinska Institute and Hospital, S-17176 Stockholm, Sweden; and §Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, 17177 Stockholm, Sweden

GENETICS. For the article "An expression signature for p53 status in human breast cancer predicts mutation status, transcriptional effects, and patient survival," by Lance D. Miller, Johanna Smeds, Joshy George, Vinsensius B. Vega, Liza Vergara, Alexander Ploner, Yudi Pawitan, Per Hall, Sigrid Klaar, Edison T. Liu, and Jonas Bergh, which appeared in issue 38, September 20, 2005, of *Proc. Natl. Acad. Sci. USA* (**102,** 13550-13555; first published September 2, 2005; 10.1073/pnas.0506230102), the breast cancer microarray data discussed in this publication have been deposited in the National Center for Biotechnology Information's Gene Expression Omnibus database (GEO, www.ncbi.nlm.nih.gov/geo/) and are accessible through GEO Series accession no. GSE3494 [NCBI GEO] .

Gene Expression Databases

Conclusion

- **GEO** is a public repository for the archiving and distribution of gene expression data
- Best resource to get microarray and Next Generation Sequencing (RNASeq) data

Medical Databases

Introduction

Informatics in health care may be called as health informatics

- It deals with the resources, devices, and methods required to optimize the acquisition, storage, retrieval, and use of information in health and biomedicine.

(wiki)

Medical Databases

Introduction

Medical databases store and provide medical information

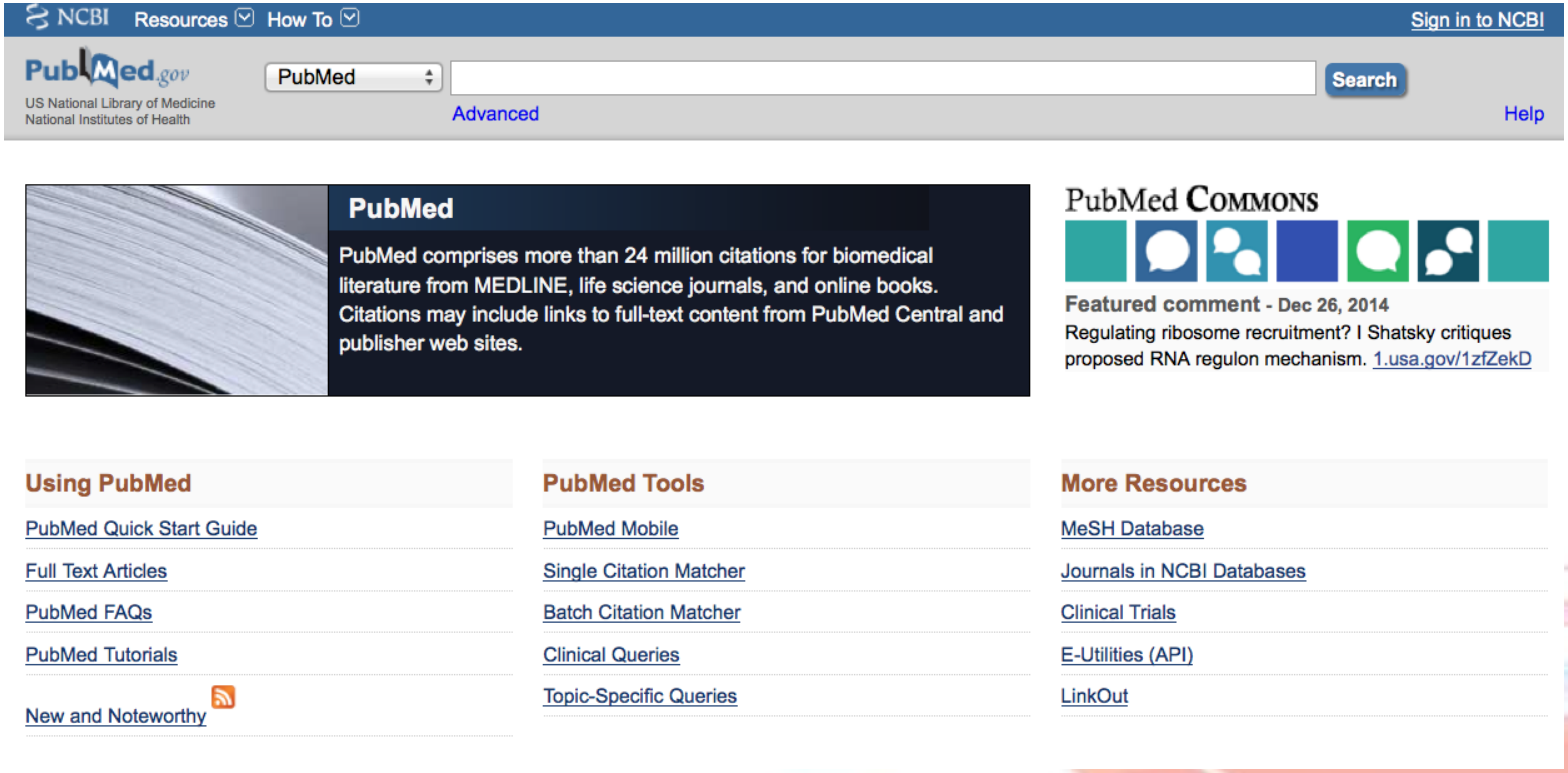
- The premier database for biomedical literature is the National Library of Medicine (NLM)'s MEDLINE, accessible through **PubMed**

Medical Databases

PUBMED

- Comprises of more than 24 million citations for biomedical literature from MEDLINE, life science journals, and online books
- Citations may include links to full-text content from PubMed Central and publisher web sites

Medical Databases



Medical Databases

MEDLINE

- MEDLINE is the primary resource for biomedical journal articles
- Millions of citations to articles in biomedical journals
- MEDLINE uses the MeSH vocabulary

Medical Databases

Other Databases

MEDLINE is the primary resource, but other databases may also be helpful

- Academic OneFile
- CINAHL (Cumulated Index of Nursing and Allied Health Literature)
- PsycINFO
- Web of Knowledge

Medical Databases

Academic OneFile

- Academic OneFile lists articles from journals covering a broad range of subjects
- While it does not primarily focus on medical topics, useful articles can still be found here

Medical Databases

EBSCO Health

Request Information

About EBSCO Health | Products | Who We Serve | Benefits | Success Stories | Contact Us

« All Products

Allied Health / Nursing

- CINAHL Complete »
- CINAHL Core eBook Packages »
- CINAHL Database »
- CINAHL Plus »
- CINAHL Plus with Full Text »
- CINAHL with Full Text »
- DynaMed »
- EDS Discovery Health »
- Health Source: Nursing/Academic Edition »
- Isabel »
- Medcom »
- Nursing & Allied Health Collection: Basic Edition »
- Nursing Reference Center »
- Nursing Reference Center Plus »
- Patient Education Reference Center »
- Rehabilitation Reference Center »
- Social Work Reference Center »

CINAHL Database

Cumulative Index to Nursing and Allied Health Literature

Nurses, allied health professionals, researchers, nurse educators and students depend on the *CINAHL Database* to research their subject area from this authoritative index of nursing and allied health journals which includes over 3 million records dating back to 1960.

Interested in learning more about CINAHL Database?

Request a Free Trial



Easy Access to the Most Authoritative Nursing and Allied Health Literature Available

CINAHL provides indexing of the top nursing and allied health literature available including nursing journals and publications from the National League for Nursing, and the American Nurses' Association. Literature covers a wide range of topics including nursing, biomedicine, health sciences librarianship, alternative/complementary medicine, consumer health and 17 allied health disciplines.

In addition, *CINAHL* provides access to health care books, nursing dissertations, selected

Content Lists

Coverage List:
PDF » | Excel » | HTML »

Content Includes

- More than 3 Million Records
- Indexing for more than 3,000 Journals
- 70 Full-Text Journals

Also Contains

- Author Affiliations
- Searchable Cited References for more than 1,250 journals

Additional Resources:

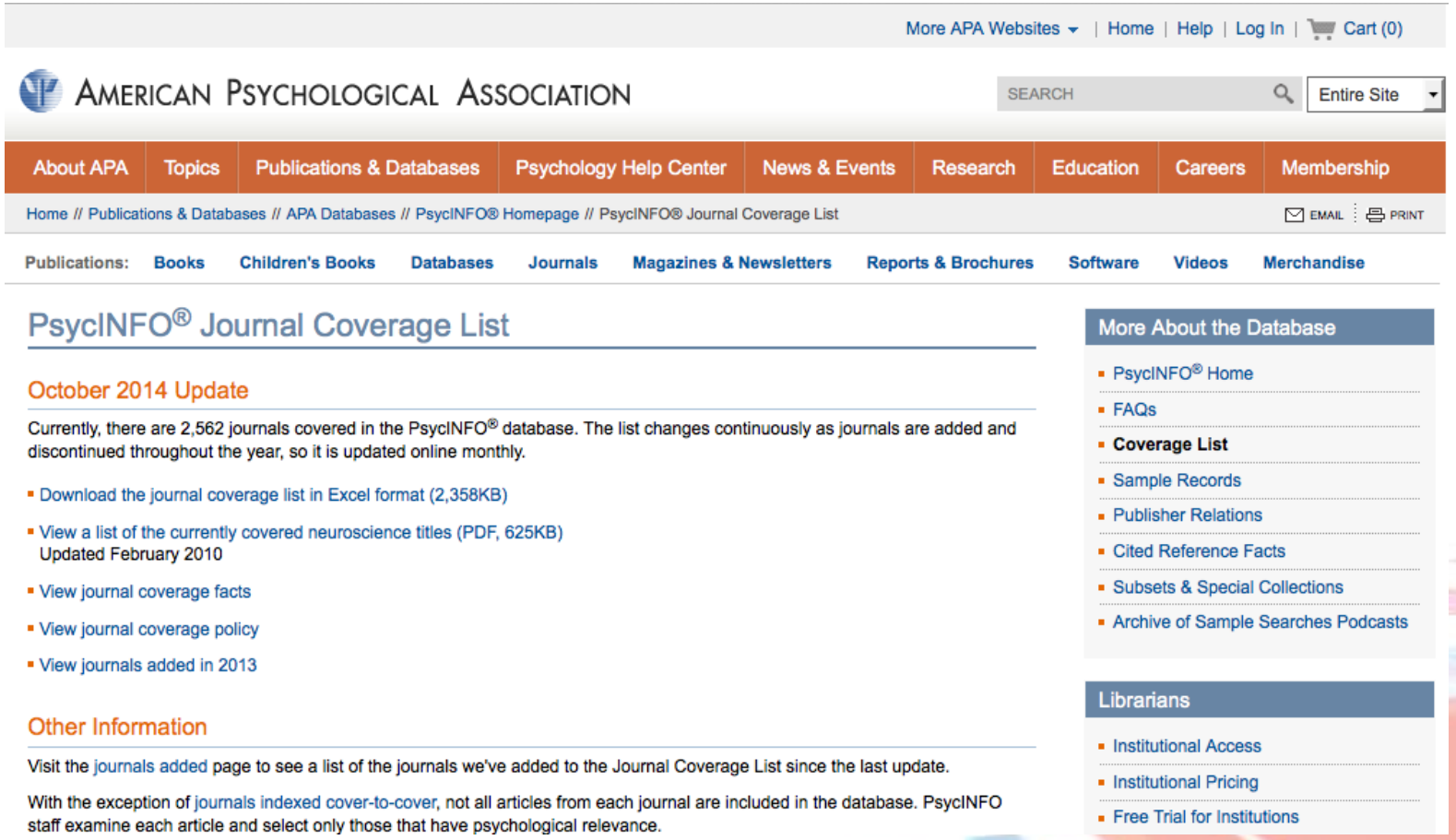
Medical Databases

PsycINFO

- PsycINFO searches the psychological literature
- While it does not primarily focus on medical topics, useful articles can still be found here

http://www.apa.org/pubs/databases/psycinfo/coverage.aspx

Medical Databases



http://www.apa.org/pubs/databases/psycinfo/coverage.aspx

Medical Databases

Web of Science

- Major source for articles in a wide range of fields, including the sciences, social sciences, and humanities.
- Excellent place to find articles from scientific journals that may not be included in MEDLINE

Medical Databases

Conclusions

Informatics in health care may be called as health informatics

- Medical databases deal with the acquisition, storage, retrieval, and use of information in health and biomedicine.

Sequence Submission

Introduction

- Sequences are submitted to the databases in order to share them with the scientific community
- Generally sequences are submitted at the time of publication and are reviewed by peers

Sequence Submission

Caution

- It is important to ensure that sequence files do not contain any special characters
- Control characters in addition to standard ASCII characters must be removed else they might mess up the analysis or data transfer

Sequence Submission

Table 2.1. *Base–nucleic acid codes*

Symbol	Meaning	Explanation
G	G	Guanine
A	A	Adenine
T	T	Thymine
C	C	Cytosine
R	A or G	puRine
Y	C or T	pYrimidine
M	A or C	aMino
K	G or T	Keto
S	C or G	Strong interactions 3 h bonds
W	A or T	Weak interactions 2 h bonds
H	A, C or T not G	H follows G in alphabet
B	C, G or T not A	B follows A in alphabet
V	A, C or G not T (not U)	V follows U in alphabet
D	A, G or T not C	D follows C in alphabet
N	A,C,G or T	Any base

Adapted from NC-IUB (1984).

Mount, pg 28

- Nomenclature committee of International Union of Biochemistry (IUB) has established standard codes to represent ambiguous bases or aminoacids

Sequence Submission

Table 2.2. *Table of standard amino acid code letters*

1-letter code	3-letter code	Amino acid
A[a]	Ala	alanine
C	Cys	cysteine
D	Asp	aspartic acid
E	Glu	glutamic acid
F	Phe	phenylalanine
G	Gly	glycine
H	His	histidine
I	Ile	isoleucine
K	Lys	lysine
L	Leu	leucine
M	Met	methionine
N	Asn	asparagine
P	Pro	proline
Q	Gln	glutamine
R	Arg	arginine
S	Ser	serine
T	Thr	threonine
V	Val	valine
W	Trp	tryptophan
X	Xxx	undetermined amino acid
Y	Tyr	tyrosine
Z[b]	Glx	either glutamic acid or glutamine

Adapted from IUPAC-IUB (1969, 1972, 1983).

[a] Letters not shown are not commonly used.

[b] Note that sometimes when computer programs translate DNA sequences, they will put a "Z" at the end to indicate the termination codon. This character should be deleted from the sequence.

Mount, pg 28

Sequence Submission

NCBI

NCBI has two options

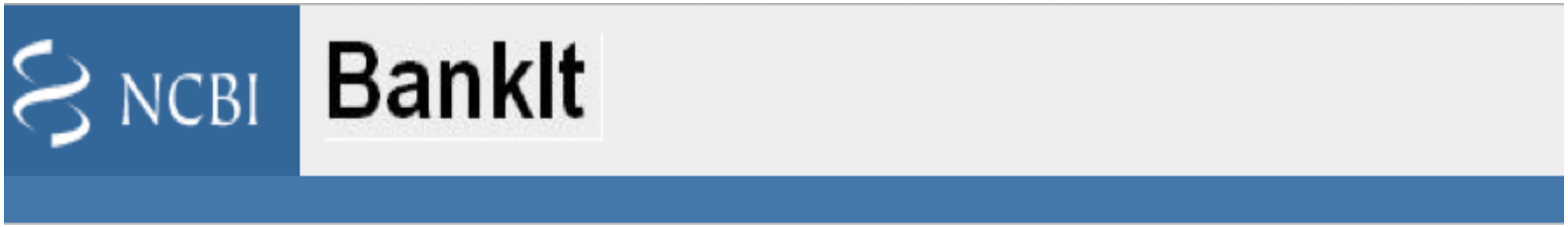
- **BANKIt**
  - For simple sequences and annotations
  - For submission through web
  - Do not require advanced tools

Sequence Submission

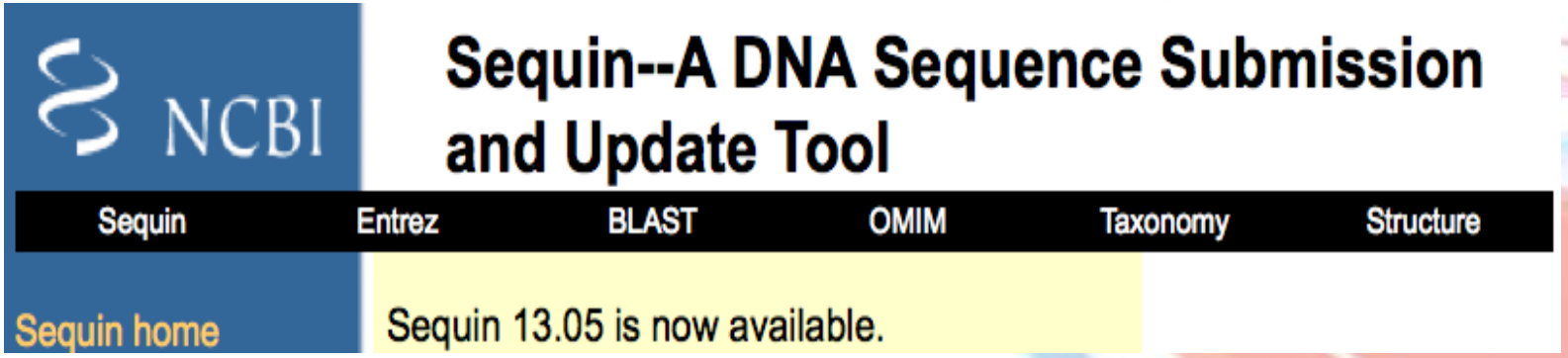
NCBI

- **Sequin**
  - For Complex sequences and annotations
  - For off-line submissions
  - Do require advanced tools and graphical reports

Sequence Submission



http://www.ncbi.nlm.nih.gov/WebSub/?tool=genbank



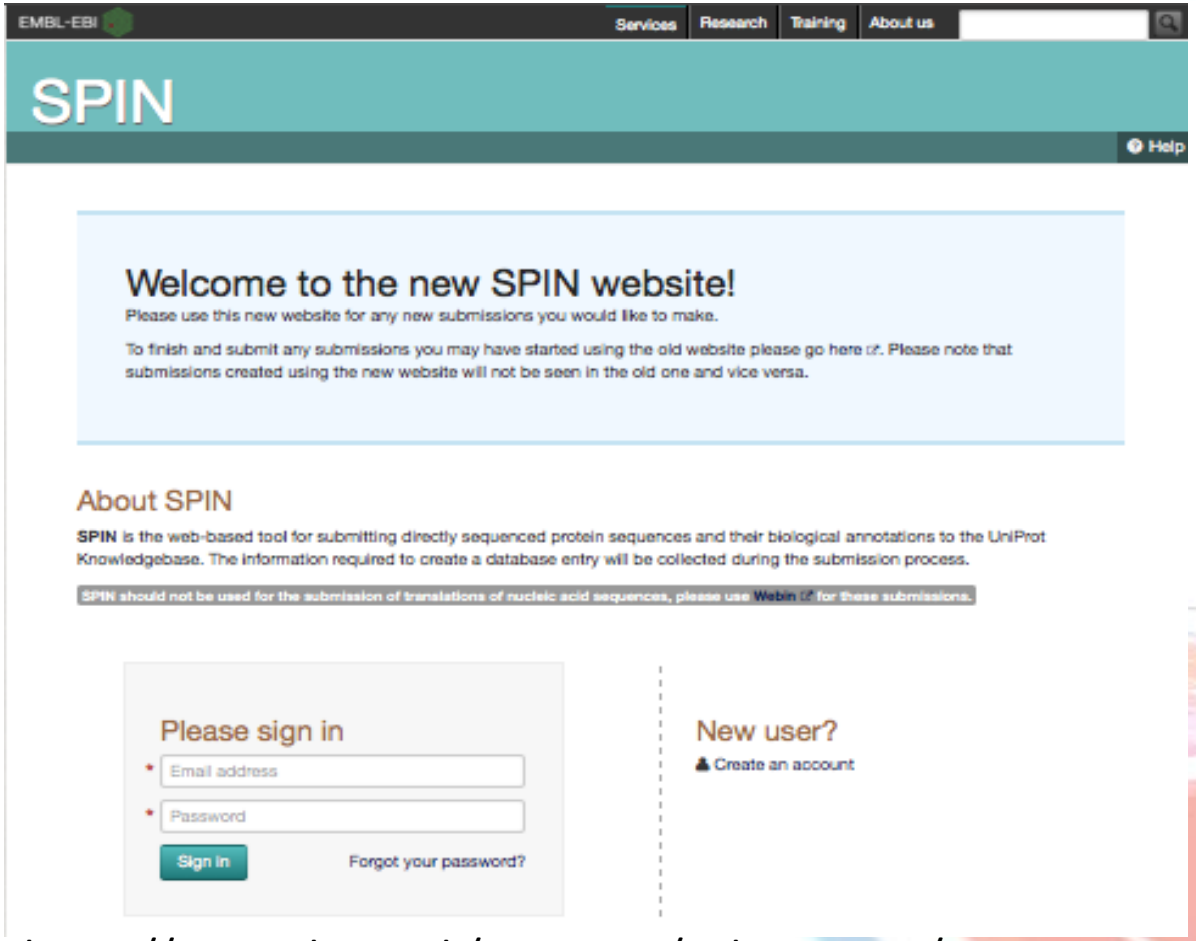
Sequence Submission

UniProt

SPIN web-based tool for submitting

- directly sequenced protein sequences
- biological annotations to the knowledgebase

Sequence Submission



https://www.ebi.ac.uk/swissprot/Submissions/spin

Sequence Submission

Conclusion

- Sequences are stored in databases in specific format



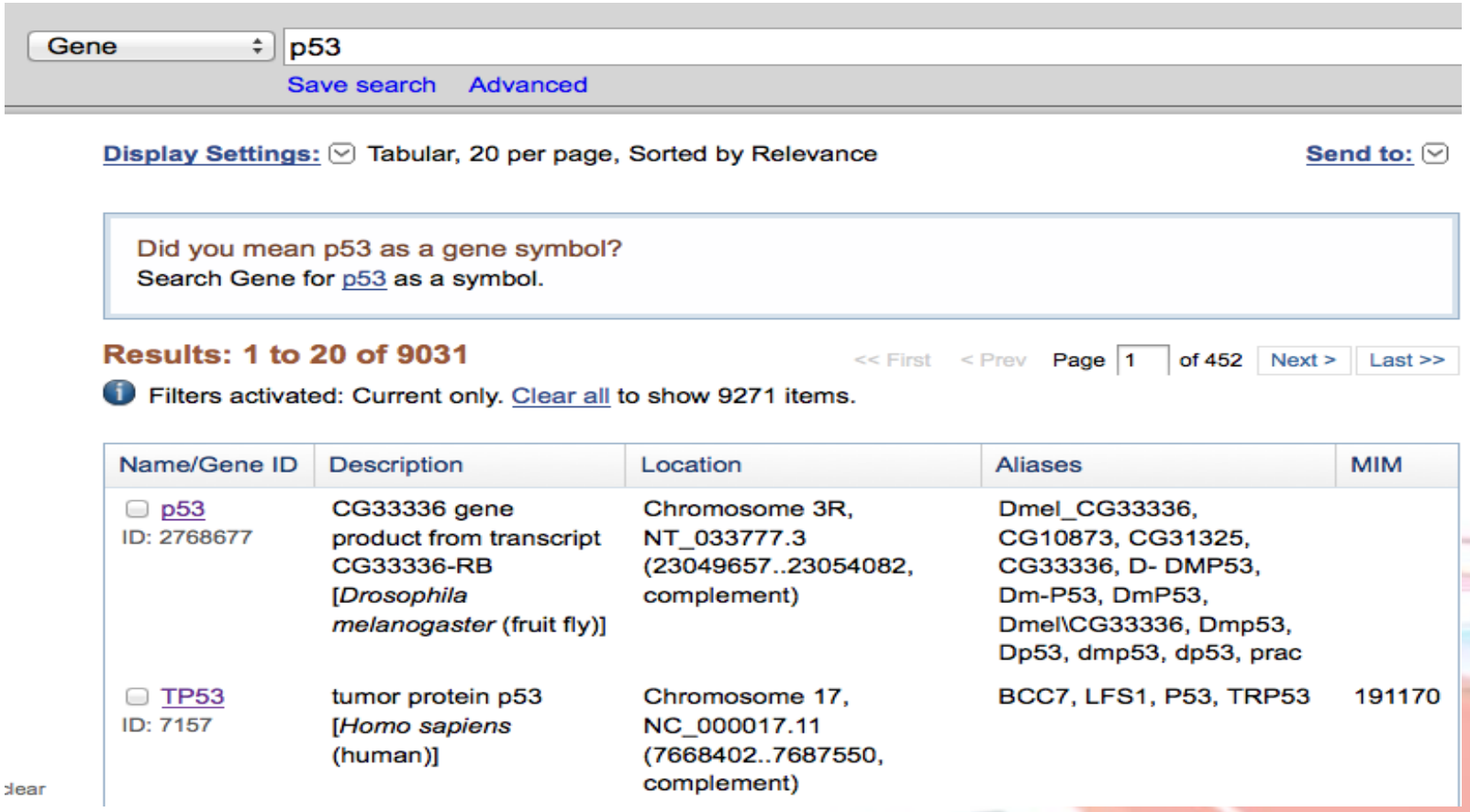
DNA Sequence Retrieval

Introduction

- Databases not merely collect and organize data but allow intelligent data retrieval
- And/or analysis



DNA Sequence Retrieval



http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

p53 [*Drosophila melanogaster* (fruit fly)]

Gene ID: 2768677, updated on 4-Jan-2015

Summary

Official Symbol	p53 provided by FlyBase
Primary source	FLYBASE:FBgn0039044
Locus tag	Dmel_CG33336
Gene type	protein coding
RefSeq status	REVIEWED
Organism	Drosophila melanogaster (old-lineage: Eukaryota; Metazoa; Arthropoda; Hexapoda; Insecta; Pterygota; Neoptera; Endopterygota; Diptera; Brachycera; Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora)
Lineage	Eukaryota; Metazoa; Ecdysozoa; Arthropoda; Hexapoda; Insecta; Pterygota; Neoptera; Endopterygota; Diptera; Brachycera; Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora
Also known as	CG10873; CG31325; CG33336; D-p53; Dm-P53; Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53; prac

http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

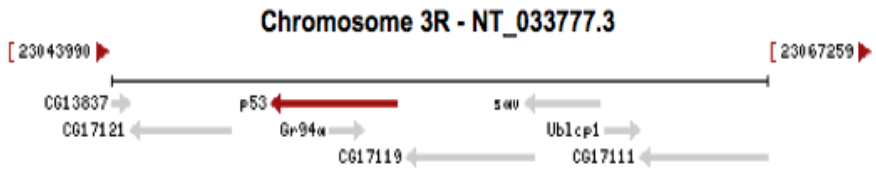
Genomic context

Location: 94D10-94D10

See p53 in Epigenomics, MapViewer

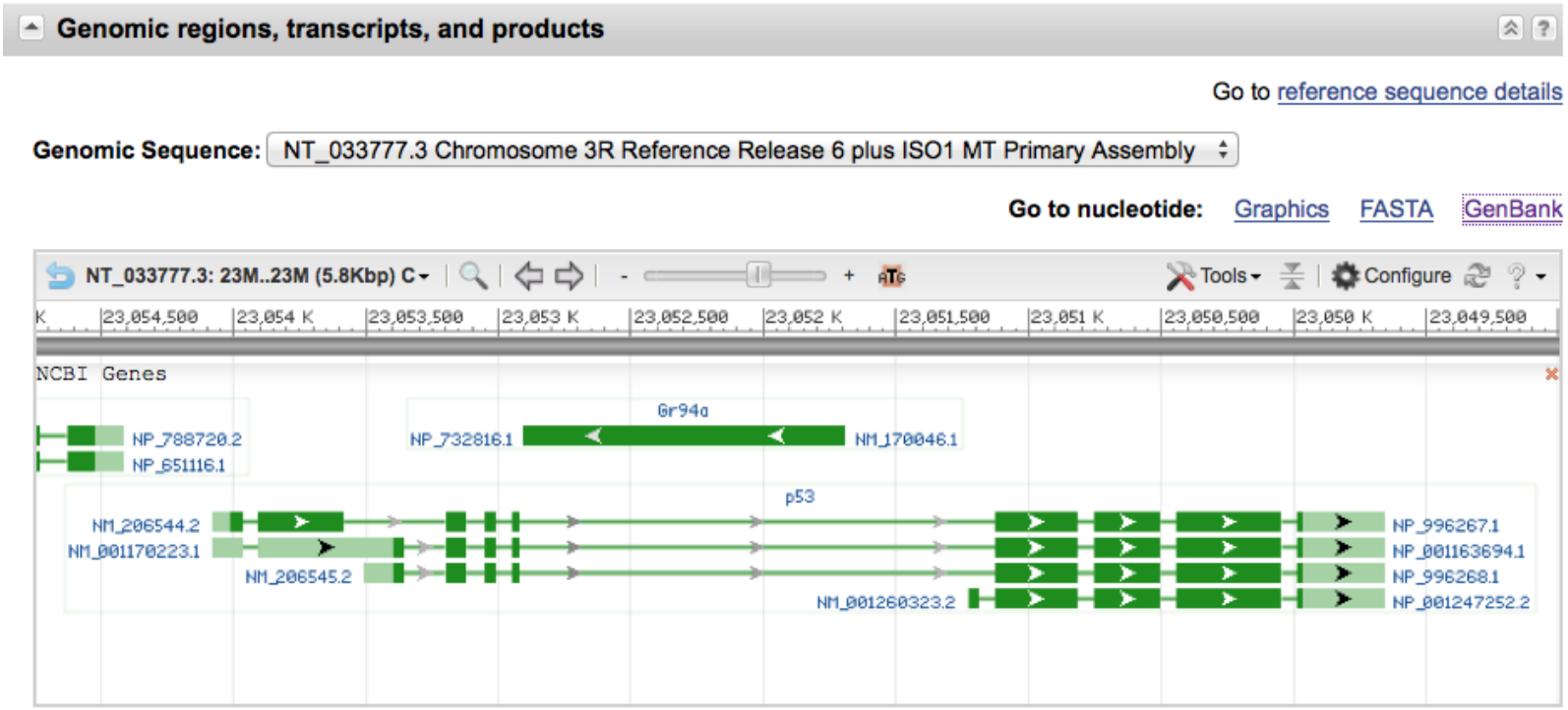
Exon count: 10

Annotation release	Status	Assembly	Chr	Location
Release 6.01	current	Release 6 plus ISO1 MT (GCF_000001215.4)	3R	NT_033777.3 (23049657..23054082, complement)
Release 5.57	previous assembly	Release 5 (GCF_000001215.2)	3R	NT_033777.2 (18875379..18879804, complement)



http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval



http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

Drosophila melanogaster chromosome 3R

NCBI Reference Sequence: NT_033777.3

FASTA Graphics

```
LOCUS       NT_033777               4426 bp    DNA     linear   INV 05-AUG-2014
DEFINITION  Drosophila melanogaster chromosome 3R.
ACCESSION   NT_033777 REGION: complement(23049657..23054082)
VERSION     NT_033777.3  GI:671162122
DBLINK      BioProject: PRJNA164
            BioSample: SAMN02803731
KEYWORDS    RefSeq.
SOURCE      Drosophila melanogaster (fruit fly)
  ORGANISM  Drosophila melanogaster
            Eukaryota; Metazoa; Ecdysozoa; Arthropoda; Hexapoda; Insecta;
            Pterygota; Neoptera; Endopterygota; Diptera; Brachycera;
            Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora.
REFERENCE   1  (bases 1 to 4426)
  AUTHORS   Hoskins,R.A., Carlson,J.W., Kennedy,C., Acevedo,D., Evans-Holm,M.,
            Frise,E., Wan,K.H., Park,S., Mendez-Lago,M., Rossi,F.,
            Villasante,A., Dimitri,P., Karpen,G.H. and Celniker,S.E.
  TITLE     Sequence finishing and mapping of Drosophila melanogaster
            heterochromatin
  JOURNAL   Science 316 (5831), 1625-1628 (2007)
   PUBMED   17569867
```

http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

```
FEATURES             Location/Qualifiers
     source          1..4426
                     /organism="Drosophila melanogaster"
                     /mol_type="genomic DNA"
                     /db_xref="taxon:7227"
                     /chromosome="3R"
                     /genotype="y[1]; Gr22b[1] Gr22d[1] cn[1] CG33964[R4.2]
                     bw[1] sp[1]; LysC[1] MstProx[1] GstD5[1] Rh6[1]"
     gene            1..4426
                     /gene="p53"
                     /locus_tag="Dmel_CG33336"
                     /gene_synonym="CG10873; CG31325; CG33336; D-p53; Dm-P53;
                     Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53;
                     prac"
                     /map="94D10-94D10"
                     /db_xref="FLYBASE:FBgn0039044"
                     /db_xref="GeneID:2768677"
     mRNA            join(1..118,178..501,884..964,1035..1071,1135..1161,
                     2959..3268,3333..3579,3642..4036,4096..4426)
                     /gene="p53"
                     /locus_tag="Dmel_CG33336"
                     /gene_synonym="CG10873; CG31325; CG33336; D-p53; Dm-P53;
                     Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53;
                     prac"
                     /product="p53, transcript variant B"
                     /note="p53-RB; Dmel\p53-RB; CG33336-RB; Dmel\CG33336-RB"
                     /transcript_id="NM_206544.2"
                     /db_xref="GI:281362333"
                     /db_xref="FLYBASE:FBtr0084360"
                     /db_xref="FLYBASE:FBgn0039044"
                     /db_xref="GeneID:2768677"
```

http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

```
CDS             join(75..118,178..501,884..964,1035..1071,1135..1161,
                2959..3268,3333..3579,3642..4036,4096..4118)
                /gene="p53"
                /locus_tag="Dmel_CG33336"
                /gene_synonym="CG10873; CG31325; CG33336; D-p53; Dm-P53;
                Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53;
                prac"
                /note="CG33336 gene product from transcript CG33336-RB;
                CG33336-PB; p53-PB; p53-like regulator of apoptosis and
                cell cycle; Dmp53; protein 53; drosophila p53"
                /codon_start=1
                /product="p53, isoform B"
                /protein_id="NP_996267.1"
                /db_xref="GI:45553461"
                /db_xref="FLYBASE:FBpp0083753"
                /db_xref="FLYBASE:FBgn0039044"
                /db_xref="GeneID:2768677"
                /translation="MSLHKSASFSLTFNQNTSIVSRSNSRTIFEAFKEFLDFWDIGNE
                VSAESAVRVSSNGAFNLPQSFGNESNEYAHLATPVDPAYGGNNTNNMMQFTNNLEILA
                NNNSDGNNKINACNKFVCHKGTDSEDDSTEVDIKEDIPKTVEVSGSELTTEPMAFLQG
                LNSGNLMQFSQQSVLREMMLQDIQIQANTLPKLENHNIGGYCFSMVLDEPPKSLWMYS
                IPLNKLYIRMNKAFNVDVQFKSKMPIQPLNLRVFLCFSNDVSAPVVRCQNHLSVEPLT
                ANNAKMRESLLRSENPNSVYCGNAQGKGISERFSVVVPLNMSRSVTRSGLTRQTLAFK
                FVCQNSCIGRKETSLVFCLEKACGDIVGQHVIHVKICTCPKRDRIQDERQLNSKKRKS
                VPEAAEEDEPSKVRRCIAIKTEDTESNDSRDCDDSAAEWNVSRTPDGDYRLAITCPNK
                EWLLQSIEGMIKEAAAEVLRNPNQENLRRHANKLLSLKKRAYELP"
```

http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

```
ORIGIN
        1 cctggagcac ggaagattct tgcggacaca aatcgcaact gctaaataaa atttatttat
       61 ttgagtgcac agccatgagt cttcacaagt ccgcgtcgtt tagcttgact tttaaccagt
      121 gagcggagat attttattcg gtcttaccca acaaaataat gttgcgcctt tttgcagaaa
      181 cacttcgatt gtttcgcgta gcaatagtcg cacaattttt gaagctttca aggagttcct
      241 ggatttttgg gatatcggca acgaagtttc tgcagagtca gcagttcggg tctccagcaa
      301 cggagctttc aacttgccgc agagttttgg caacgaatcc aacgaatatg cccacctggc
      361 tacgcctgtg gatccagcct acggaggcaa caacacgaac aacatgatgc agttcacgaa
      421 caatctggaa attttggcca acaataattc cgatggcaat aacaaaatta atgcatgcaa
      481 caaattcgtc tgccacaagg ggtgagcaaa ttcaaaacac gcgctccaat cgataaacat
      541 tggctacggc gattgttcgc gctgcgtggc gaatggcaaa atccaaatag tcggtggcca
      601 ctacgattct gtagtttttt gttagcgaat ttttaatatt tagcctcctt ccccaacaag
      661 atcgcttgat cagatatagc cgactaagat gtatatatca cagccaatgt cgtggcacaa
      721 agaaaggtac agtgcggcaa caaattgatg atcgaacagt agaaaccttg catgtagcaa
       ------------------------------------
     4261 ggcatgttcg atggccgaaa agaaaacatt tttatatttt tgatagtata ctgttgttaa
     4321 ctgcagttct atgtgactac gtaacttttg tctaccacaa caaacatact ctgtacaaaa
     4381 aagccaaaag tgaatttatt aaagagttgt catattttgc aaacat
//
```

http://www.ncbi.nlm.nih.gov/

DNA Sequence Retrieval

Conclusions

- DNA Sequences are stored in DNA sequence databases in specified formats
- Genebank format is a standard format

Lecture no.10

Protein Sequence Retrieval

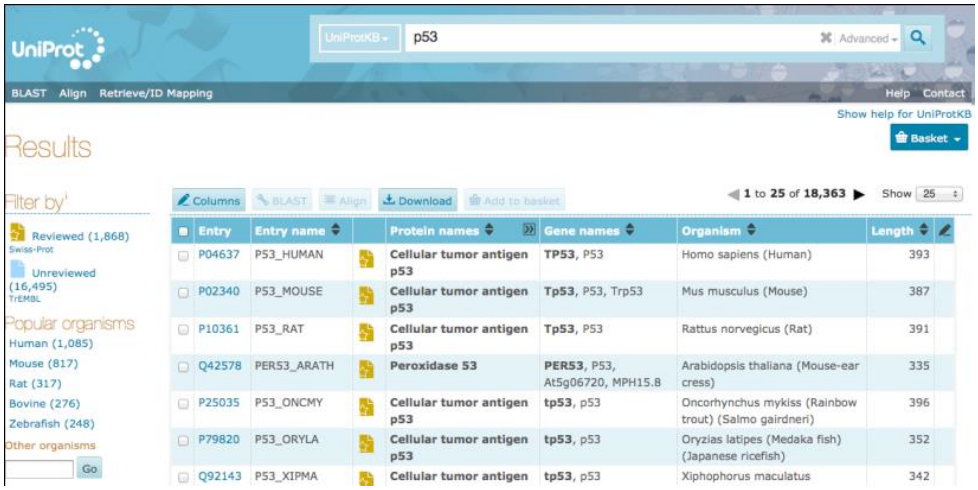
Protein Sequences:

Now we talk about the data retrieval and first we'll talk about the protein sequence retrievals and structures. Protein data is of the following types:

- Actual sequences (from the proteomic data or some other experimental techniques) or translated sequences (sometimes, we go to nucleotides databases, we get those nucleotides and then we translate them by using some softwares, so these are kind of predicted protein sequences) .
- Structures (we can also make structures from those proteins that maybe predicted or the real structures coming from various X-ray Crystallography Techniques).
- Annotations (sometimes, we are interested in the functions of the proteins so those are stored as annotations).

UniProt (*It is an international partnership between PIR, EBI and SIB*):

Now as far as the resources are concerned, we have multiple resources for protein sequences but **UniProt**claims to be the biggest and integrated resource whereas for the structures **PDB** seems like a good resource.



As shown in his figure, is the webpage for data retrieval from UniProt, so we want to search a protein, say p53, where we put it into the search box and press enter which gives us the output. And we see that there are

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18,000 different records and it is showing us the first 25 out of them.

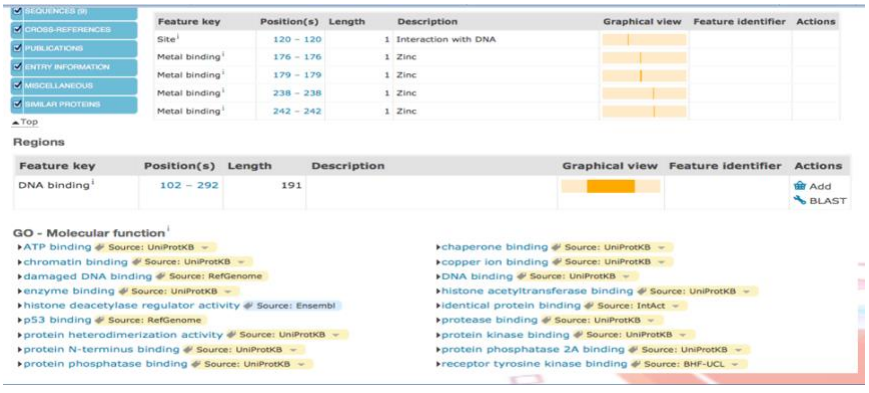
We can have different columns for the output on this webpage so we can have *entry; it's ID, entry name* (the Suffix Human is written so it's coming from Human, it can be from mouse, rat and Arabidopsis), the *protein name* is Cellular tumour antigen, then *gene name* which is TP53 (where TP stands for Tumour Protein), the *organism* is obviously the human (here) and in the end we have it's length i.e. 393 bp (base pairs).

The link to this webpage is **http://www.uniprot.org/uniprot/.**



So, let's check the first one and here we reach on the record for this protein (shown in the figure on the left) which is Cellular Tumour Antigen p53 protein, commonly known as TP53.

We can have different tabs, showing us the outputs. We can look into the *functions*, its *taxonomy*, and lot many other characteristics so if we look into the function so it gives us some description about what it's doing.



After scrolling the same webpage (shown in the figure on the left), we can see the *feature key* and in some *site* written (there are unique sites in different proteins which are having some specific properties in them so this is just one amino-acid present in this protein that *interacts with the DNA*). Similarly, there are different *metal binding sites* and we can see that it's

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mainly binding to the *Zinc*metal.The number of amino-acids is shown here so these are the regions where it interacts with the metal.

Down below, we can also see the *DNA binding* region, for example here, the amino acids are from 102 to 292 and that is also shown in the *Graphical view* as well.

GO-Molecular function or GO-Gene Ontologies, so gene ontologies are the different functional annotation term, there they define different functions, so amongst them we have molecular functions, biological processes, and we have cellular components. So here we just see a *Molecular function,* so it tells us that it performs the functions as shown in the figure , mainly it's a *ATP binding, it's p53 binding* with various other functions like *DNA binding.* So all those functions related to these proteins are present in the heading of GO-Molecular Function.



Next, we move on to some other functions, in the *Biological process* category (shown in the figure) we see that it is related to *Apoptosis* (which is a cell death and it is related to cell-cycle and some other components).

In the below section, we see some *enzymes and pathway databases,* and*Reactome*is a database in which we have a group of reactions which are categorized so these are the list of those reactions with which it is related.

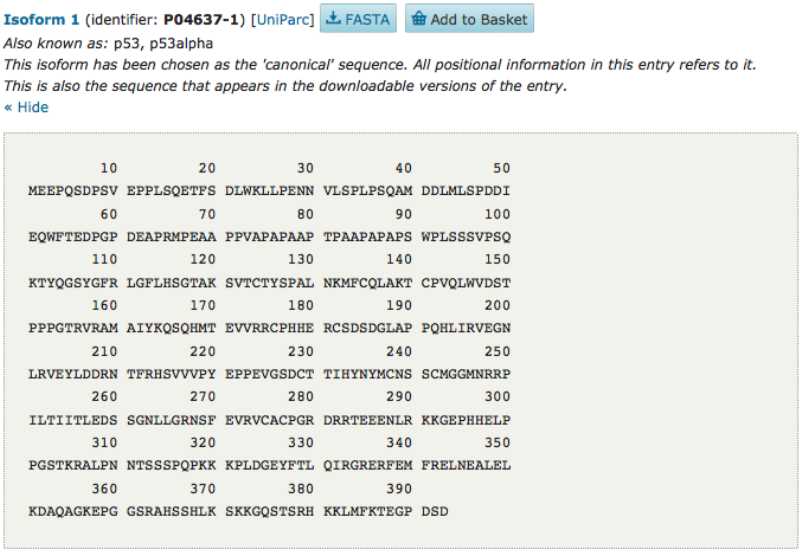


When we move further (as shown in the figure on the left) till we reach its *Taxonomy.*

On the top, we can see something written as *Protein family* or *group databases* whichis TCDB. Basically, there is another classification in which the proteins are classified on the basis of being as transporter proteins so it is associated with the transportation across the membranes and there is 5-digit number, so there is a specific classification code which is given to each protein, and this protein has the specific code as shown in the figure.

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So then we have the *names and taxonomies,* where there are *protein names,* and thetaxonomyof the individual can be seen in the *Taxonomic lineage* row. Let's see how we reach to its sequence and is shown in the figure below:



Isoform 1 (identifier: **P04637-1**) [UniParc] FASTA Add to Basket

Also known as: p53, p53alpha

This isoform has been chosen as the 'canonical' sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

« Hide

```
        10         20         30         40         50
MEEPQSDPSV EPPLSQETFS DLWKLLPENN VLSPLPSQAM DDLMLSPDDI
        60         70         80         90        100
EQWFTEDPGP DEAPRMPEAA PPVAPAPAAP TPAAPAPAPS WPLSSSVPSQ
       110        120        130        140        150
KTYQGSYGFR LGFLHSGTAK SVTCTYSPAL NKMFCQLAKT CPVQLWVDST
       160        170        180        190        200
PPPGTRVRAM AIYKQSQHMT EVVRRCPHHE RCSDSDGLAP PQHLIRVEGN
       210        220        230        240        250
LRVEYLDDRN TFRHSVVVPY EPPEVGSDCT TIHYNYMCNS SCMGGMNRRP
       260        270        280        290        300
ILTIITLEDS SGNLLGRNSF EVRVCACPGR DRRTEEENLR KKGEPHHELP
       310        320        330        340        350
PGSTKRALPN NTSSSPQPKK KPLDGEYFTL QIRGRERFEM FRELNEALEL
       360        370        380        390
KDAQAGKEPG GSRAHSSHLK SKKGQSTSRH KKLMFKTEGP DSD
```

In this figure, we can see the sequence of the protein which is found to be at the end of the page.

Here, it says *Isoform 1,* so different proteins have different isoforms, different alternative splice variants so this is Isoform 1 as exhibited by its name which is P04637-1, and is the kind of first isoform. We can see the sequence of the protein and starts with a methionine (always a first amino acid in those proteins) and ending at 390TH amino acid. So, it's a 393 aa long protein and the sequence is right here. You can click on the FASTA button on the top and then you can get this output in FASTA format (we'll discuss it later).

NCBI:

```
ORIGIN
        1 meepqsdpsv epplsqetfs dlwkllpenn vlsplpsqam ddlmlspddi eqwftedpgp
       61 deaprmpeaa ppvapapaap tpaapapaps wplsssvpsq ktyqgsygfr lgflhsgtak
      121 svtctyspal nkmfcqlakt cpvqlwvdst pppgtrvram aiykqsqhmt evvrrcphhe
      181 rcsdsdglap pqhlirvegn lrveylddrn tfrhsvvvpy eppevgsdct tihynymcns
      241 scmggmnrrp iltiitleds sgnllgrnsf evrvcacpgr drrteeenlr kkgephhelp
      301 pgstkralpn ntsssppkk kpldgeyftl qirgrerfem frelnealel kdaqagkepg
      361 gsrahsshlk skkgqstsrh kklmfktegp dsd
//
```

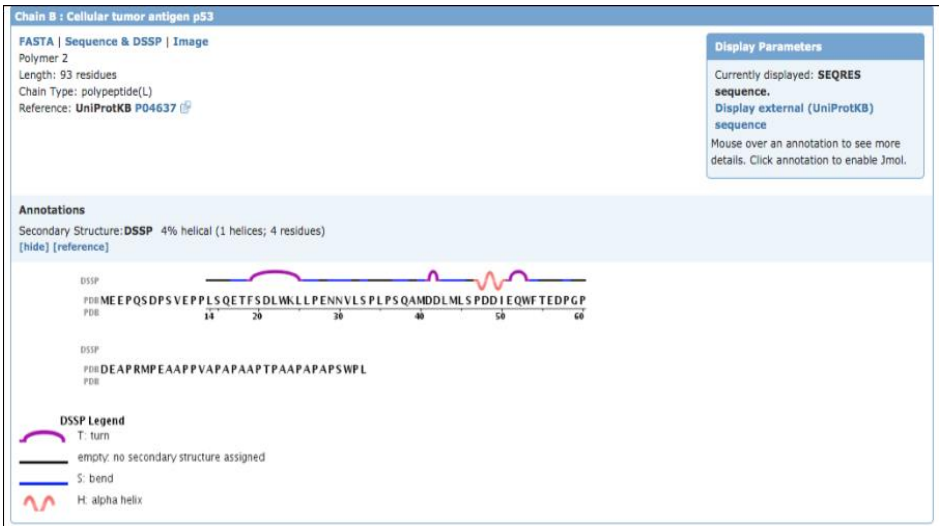
We can also get the same protein from NCBI (as shown in the figure on the left)

In NCBI, obviously the sequence is pretty similar and the arrangement is slightly different so it is *ORIGIN,* where the sequence starts and sequence ends at those two

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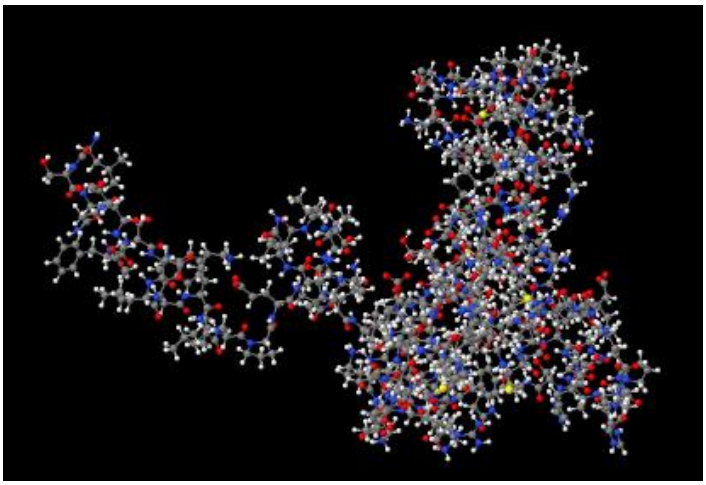
slashes (//). So, we can get the protein sequence from NCBI as well and the link to this website is http://www.ncbi.nlm.nih.gov/.

PDB:



PDB gives us the structures, so we can go to PDB webpage (as shown in the figure on the left) and search for the same ID i.e. P04637 and it gives us the sections or the regions from where it can make up some specific structures.

You can see the *turns* in Annotations section, the black ones are the empty lines where no secondary structure can be formed, blue ones show those bends and the orange ones are designated as alpha helices regions. So in PDB, we can have structures in this format as well as the 3D-Structures as shown in the figure below:



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The link towards PDB is:
http://www.rcsb.org/pdb/explore/remediatedSequence.do?structureId=2LY4&bionumber=1

Conclusions:

We conclude that:

- UniProt is the integrated resource between PIR, EBI and SIB and
- PDB is a good resource to get the protein structure.

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Sequence Formats

FASTA Sequence Format

- DNA sequences are stored in specified formats
- Different softwares need sequences in different formats

Sequence Formats

FASTA Sequence Format

- FASTA is the most frequently used format to present DNA and Protein sequences
- It is recommended that all lines of text be shorter than 80 characters in length

Sequence Formats

>gi|568815581:c7687550-7668402 Homo sapiens chromosome 17, GRCh38 Primary Assembly
GATGGGATTGGGGTTTTCCCCTCCCATGTGCT
CATCTAGAGCCACCGTCCAGGGAGCAGGTAGC
TGCTGGGCTCTCCACGACGGTGACACGC--------

>gi|120407068|ref|NP_000537.3| cellular tumor antigen p53 isoform a [Homo sapiens]
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLS
PLAPPVLGFLHSGTAKSVTCTYSPALNKMFCQLA
KT--*

Sometimes a * might be present in the end

Sequence Formats

Genebank Format

- A sequence file in GenBank format can contain several sequences
- One sequence starts with a line containing the word LOCUS and a number of annotation lines

Sequence Formats

Genebank Format

- The start of the sequence is marked by a line containing "ORIGIN" and the end of the sequence is marked by two slashes ("//")

Sequence Formats

Genebank Format

LOCUS AAU03518 237 bp DNA PRI 04-FEB-1995
DEFINITION Aspergillus awamori internal transcribed spacer 1 (ITS1) and 18S rRNA and 5.8S rRNA genes, partial sequence.
ACCESSION U03518
BASE COUNT 41 a 77 c 67 g 52 t
ORIGIN
1 aacctgcgga aggatcatta ccgagtgcgg gtcctttggg cccaacctcc catccgtgtc
61 tattgtaccc tgttgcttcg gcgggcccgc cgcttgtcgg ccgccggggg ggcgcctctg
121 ccccccgggc ccgtgcccgc cggagacccc aacacgaaca ctgtctgaaa gcgtgcagtc
181 tgagttgatt gaatgcaatc agttaaaact ttcaacaatg gatctcttgg ttccggc
//

Sequence Formats

EMBLFormat

- An example sequence in EMBL format is:

```
ID   AA03518    standard; DNA; FUN; 237 BP.
XX
AC   U03518;
XX
DE   Aspergillus awamori internal transcribed spacer 1 (ITS1) and 18S
DE   rRNA and 5.8S rRNA genes, partial sequence.
XX
SQ   Sequence 237 BP; 41 A; 77 C; 67 G; 52 T; 0 other;
     aacctgcgga aggatcatta ccgagtgcgg gtcctttggg cccaacctcc catccgtgtc      60
     tattgtaccc tgttgcttcg gcgggcccgc cgcttgtcgg ccgccggggg ggcgcctctg     120
     ccccccgggc ccgtgcccgc cggagacccc aacacgaaca ctgtctgaaa gcgtgcagtc     180
     tgagttgatt gaatgcaatc agttaaaact ttcaacaatg gatctcttgg ttccggc        237
//
```

Sequence Formats

SwissProt Format

- SwissProt protein sequence format is similar to EMBL format
- There is considerably more information about physical and biochemical properties of a protein is provided

Sequence Formats

SwissProt Format

ID - Identification.

AC - Accession number(s).
DT - Date.
DE - Description.
GN - Gene name(s).
OS - Organism species.
OG - Organelle.
OC - Organism classification.
RN - Reference number.

RP - Reference position.
RC - Reference comments.
RX - Reference cross-references.
RA - Reference authors.
RL - Reference location.
CC - Comments or notes.
DR - Database cross-references.
KW - Keywords.
FT - Feature table data.
SQ - Sequence header.
// - Termination line.

Sequence Formats

XML Format

- Extensible Markup Language
- Readable by both man and machine
- Becoming standard data format for transferring genome data

Sequence Formats

XML Format

```
<xsd:annotation>
    <xsd:documentation>
      XML Schema for SBOL core data model
compatible with RDF/XML serialization.
      <dc:date>2012-01-19</dc:date>
      <dc:creator>Evren Sirin</dc:creator>
      <dc:contributor>Michal
Galdzicki</dc:contributor>
    </xsd:documentation>
  </xsd:annotation>
```

Sequence Formats

Sequence converters

- READSEQ is a useful sequence converter developed by D.G.Gilbert at Indiana University, USA
- Recognizes DNA or Protein sequence file and interconvert them

Sequence Formats

Conclusions

- Databases store sequences in specified formats
- Genebank, DDBJ and EMBL has similar formats
- Different softwares need sequences in different formats

Data Retrieval

Data Retrieval

- Nearly all biological databases are available for download as simple text (flat) files
- A local version of the database allows one greater freedom in processing the data

Data Retrieval

Entrez

- is an integrated search engine which allows users to search and retrieve different data from the NCBI
- It can be accessed from the site www.ncbi.nlm.nih.gov/Entrez/

Data Retrieval

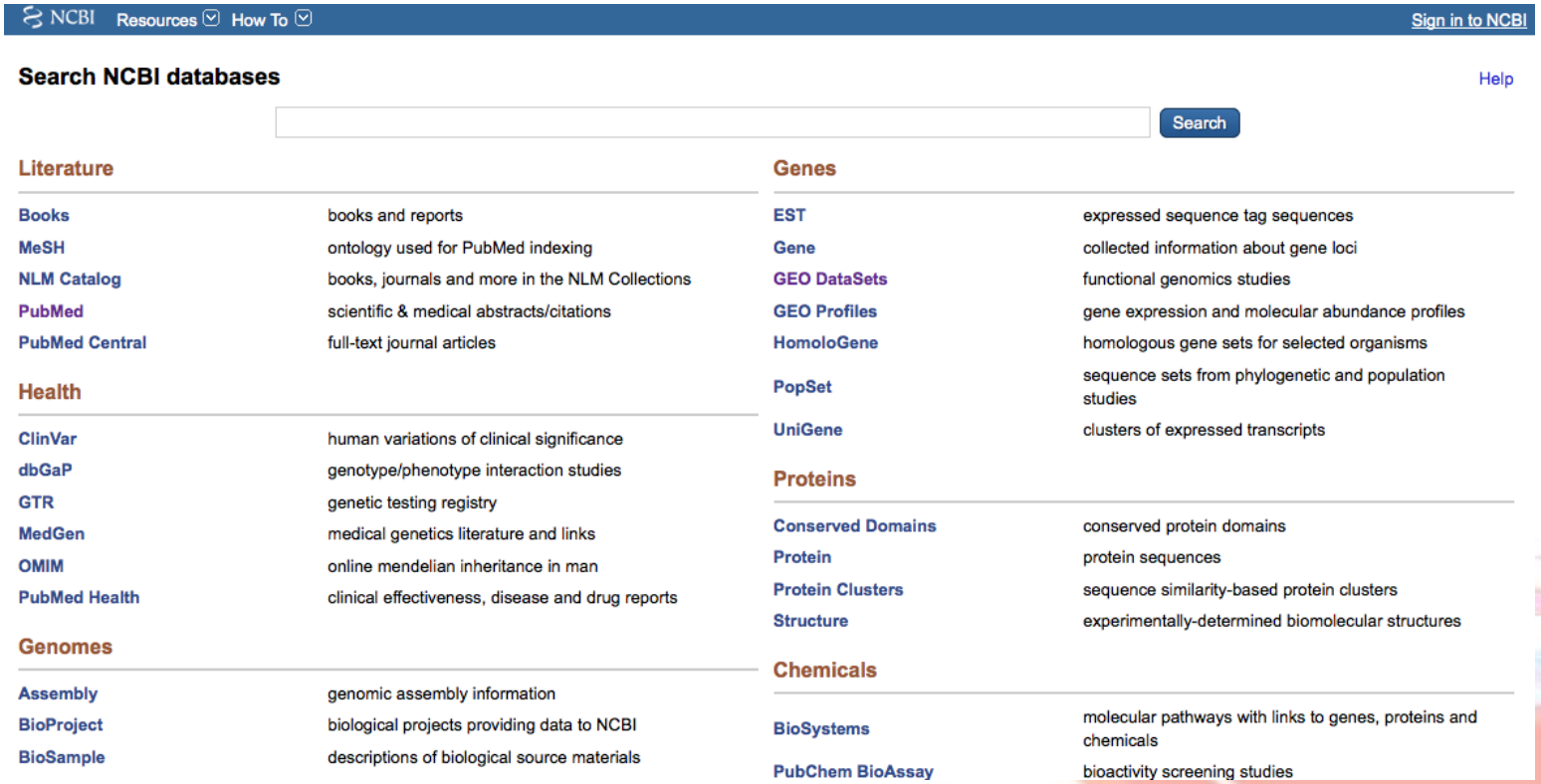
Entrez

- integrates PubMed and 39 other scientific literatures, nucleotide and protein databases
- protein domain data, population studies, expression data, pathways, genome details and taxonomic information

Data Retrieval



Data Retrieval



Data Retrieval

Bulk Data Retrieval

- The best option is to use ftp (File transfer protocol)
- The File Transfer Protocol (**FTP**) is a standard network protocol used to transfer files
- Via command line or application programs like FTP clients

Data Retrieval

Bulk Data Retrieval

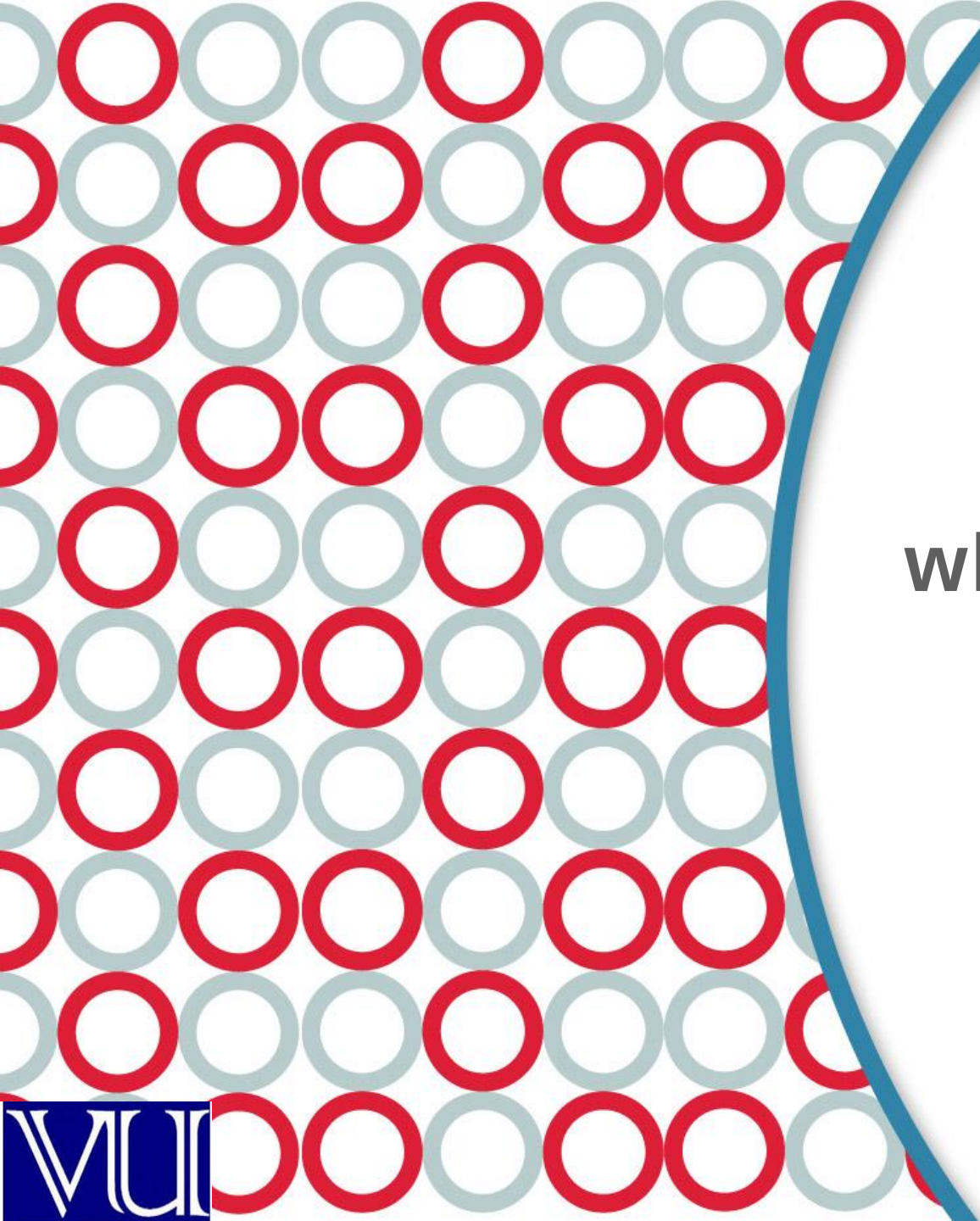
- Data needs to be transformed or processed using programming languages
- PERL and Python are good for processing Biological data

Data Retrieval

Conclusions

- Data is transferred over the internet
- Data needs to be transformed or processed





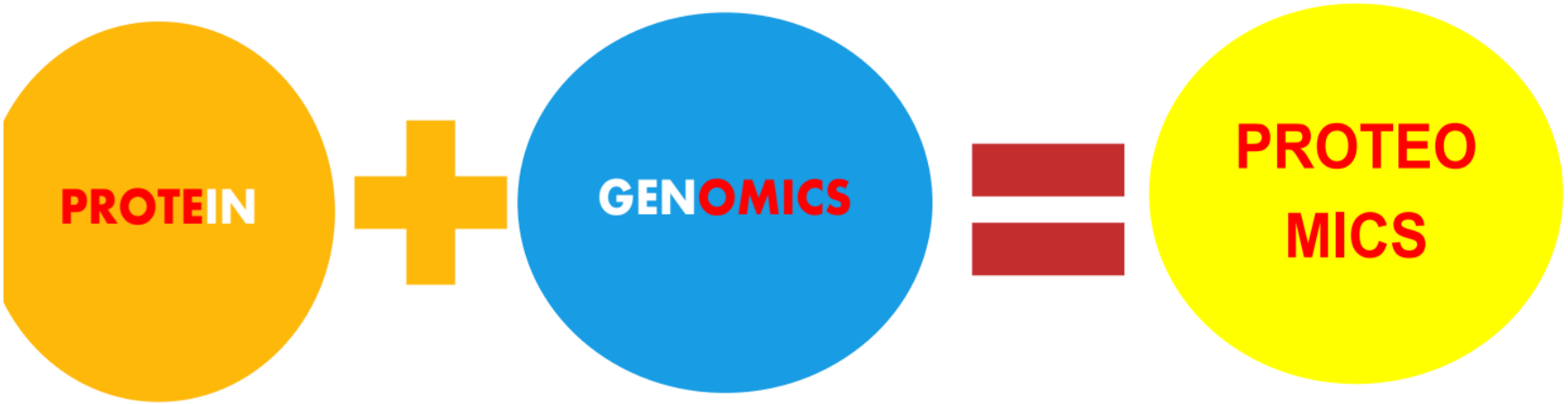
why Proteomics

Lecture-2

What do mean by proteomics?

- Proteomics is the large-scale study of proteins, usually by biochemical methods. The word proteomics has been associated
- traditionally with displaying a large number of proteins from a given
- cell line or organism on two -dimensional polyacrylamide gels

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Scope of Proteomics

The identification, characterization and quantification of all proteins involved in a particular pathway, organelle, cell, tissue, organ or organism that can be studied in concert to provide accurate and comprehensive data about that system.

Or - A complete description of proteins expressed in any given cell at any given time

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Why should we study Proteomics?

- directly contributes to drug development
- Verification of a gene product by proteomic methods
- Modifications of the proteins
- Protein expression level d

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WHY PROTEOMICS?

Many types of information cannot be obtained from the study of genes alone. For example, proteins, not genes, are responsible for the phenotypes of cells. It is impossible to elucidate mechanisms of disease, aging, and effects of the environment solely by studying the genome.

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Advantages of study of proteomics

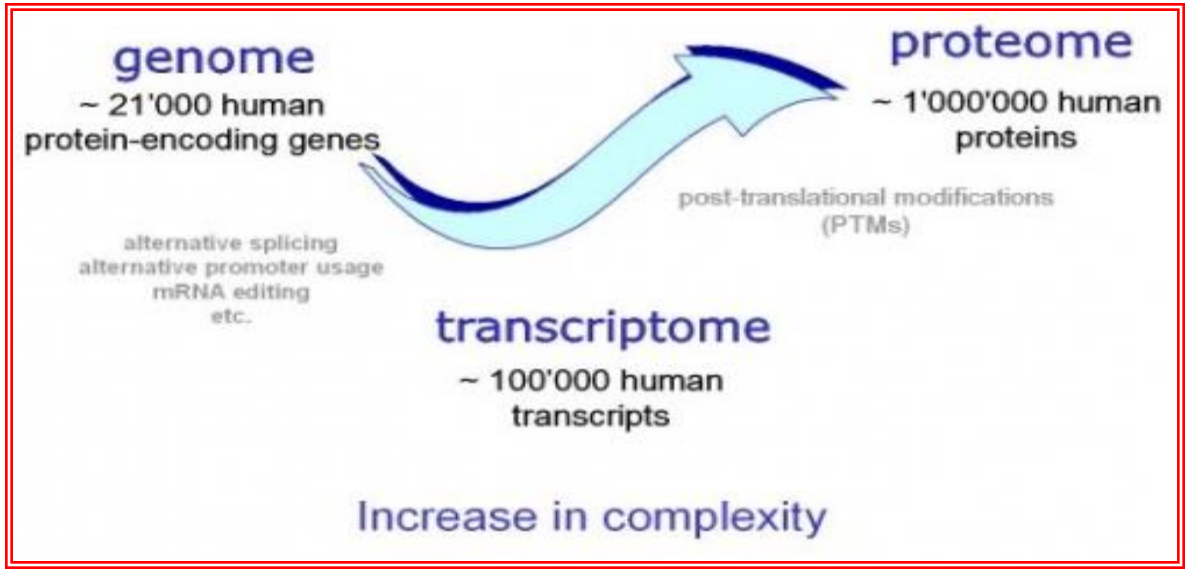
- Shows that genetic alterations are not the reason for all types of diseases
- Helps in determining the proper treatment of diseases
- With the help of three dimensional analysis of proteins we have found that HIV protease is the enzyme which is responsible for AIDS.
- One of the most important use of proteomics in diagnosis is the identification of biomarkers. The study of drugs in proteomics is called pharmacoproteomics.

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Proteomics aims

- Genomics integrated strategies
- Study of post-translational modifications
- Identification of novel protein targets for drugs
- Analysis of tumor tissues
- Comparison between normal and diseased tissues
- Comparison between diseased and pharmacologically treated tissues

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genome
~ 21'000 human protein-encoding genes
proteome
~ 1'000'000 human proteins
post-translational modifications (PTMs)
alternative splicing
alternative promoter usage
mRNA editing
etc.
transcriptome
~ 100'000 human transcripts
Increase in complexity

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Limitations of Genomics Challenge of Proteomics

- co-translational modifications
  - differential mRNA splicing
- post-translational modifications (PTMs)
  - C-terminal GPI anchor
  - phosphorylation
  - sulfation
  - glycosylation
  - N-myristoylation
  - hydroxylation
  - N-methylation
  - carboxymethylation
  - signal peptidase site.....

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